Design Journal 2

DESN1000 - Week 9

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Page Count: 73

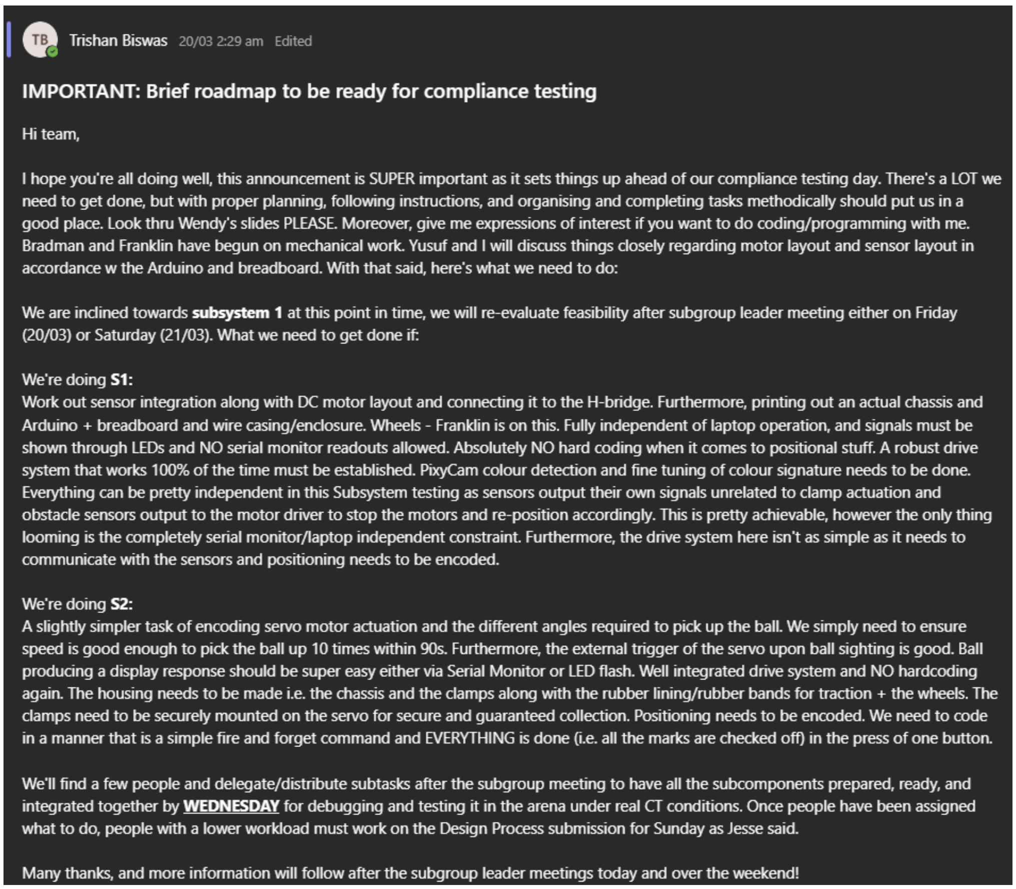
Contents

1 Design Process

1.1 Week 5

1.1.1 Entry 1 (17/03)

With the EDP presentation behind us, the team's focus now shifted to preparing for **Compliance Testing (CT)** on Week 7 Monday. On the 16th of March, we had a team meeting where Jesse outlined the priorities heading into Flexi Week. Then on the 17th, **I posted a detailed roadmap** on Microsoft Teams breaking down what we need to get done to be ready for compliance testing. See **Figure** 1 for the Teams post.



TB Trishan Biswas 20/03 2:29 am Edited

IMPORTANT: Brief roadmap to be ready for compliance testing

Hi team,

I hope you're all doing well, this announcement is SUPER important as it sets things up ahead of our compliance testing day. There's a LOT we need to get done, but with proper planning, following instructions, and organising and completing tasks methodically should put us in a good place. Look thru Wendy's slides PLEASE. Moreover, give me expressions of interest if you want to do coding/programming with me. Bradman and Franklin have begun on mechanical work. Yusuf and I will discuss things closely regarding motor layout and sensor layout in accordance w the Arduino and breadboard. With that said, here's what we need to do:

We are inclined towards **subsystem 1** at this point in time, we will re-evaluate feasibility after subgroup leader meeting either on Friday (20/03) or Saturday (21/03). What we need to get done if:

We're doing **S1:**
Work out sensor integration along with DC motor layout and connecting it to the H-bridge. Furthermore, printing out an actual chassis and Arduino + breadboard and wire casing/enclosure. Wheels - Franklin is on this. Fully independent of laptop operation, and signals must be shown through LEDs and NO serial monitor readouts allowed. Absolutely NO hard coding when it comes to positional stuff. A robust drive system that works 100% of the time must be established. PixyCam colour detection and fine tuning of colour signature needs to be done. Everything can be pretty independent in this Subsystem testing as sensors output their own signals unrelated to clamp actuation and obstacle sensors output to the motor driver to stop the motors and re-position accordingly. This is pretty achievable, however the only thing looming is the completely serial monitor/laptop independent constraint. Furthermore, the drive system here isn't as simple as it needs to communicate with the sensors and positioning needs to be encoded.

We're doing **S2:**
A slightly simpler task of encoding servo motor actuation and the different angles required to pick up the ball. We simply need to ensure speed is good enough to pick the ball up 10 times within 90s. Furthermore, the external trigger of the servo upon ball sighting is good. Ball producing a display response should be super easy either via Serial Monitor or LED flash. Well integrated drive system and NO hardcoding again. The housing needs to be made i.e. the chassis and the clamps along with the rubber lining/rubber bands for traction + the wheels. The clamps need to be securely mounted on the servo for secure and guaranteed collection. Positioning needs to be encoded. We need to code in a manner that is a simple fire and forget command and EVERYTHING is done (i.e. all the marks are checked off) in the press of one button.

We'll find a few people and delegate/distribute subtasks after the subgroup meeting to have all the subcomponents prepared, ready, and integrated together by **WEDNESDAY** for debugging and testing it in the arena under real CT conditions. Once people have been assigned what to do, people with a lower workload must work on the Design Process submission for Sunday as Jesse said.

Many thanks, and more information will follow after the subgroup leader meetings today and over the weekend!

Figure 1: My compliance testing preparation roadmap posted on Teams. **Posted by me**

The main thing I did for this entry was evaluate which subsystem set (S1 or S2) would be most realistic for us to demonstrate at compliance testing. After the team meeting, I went home and **made a flowchart** comparing the two options against where we're currently at, what resources we have, and the CT marking criteria. See **Figure** 2.

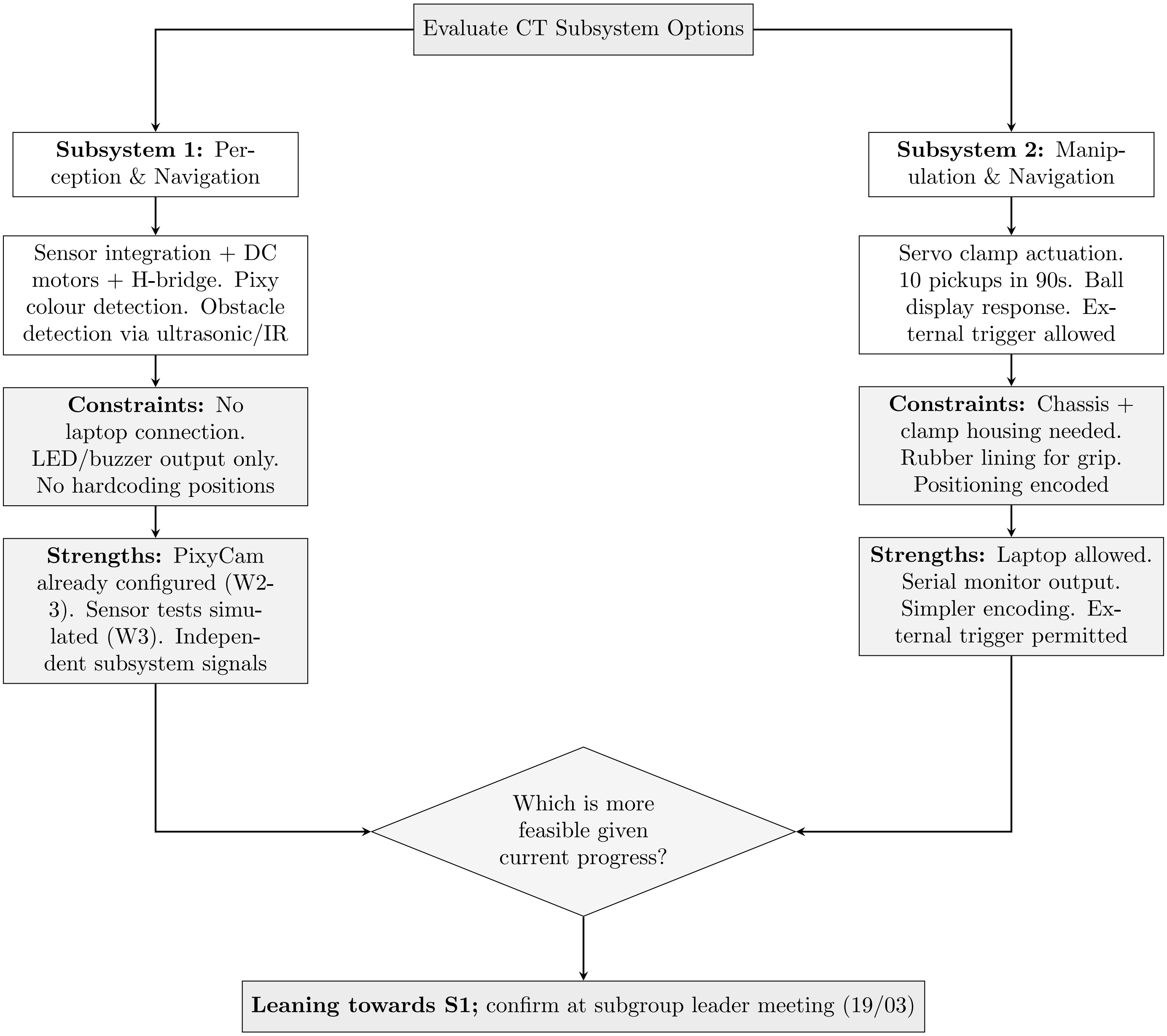


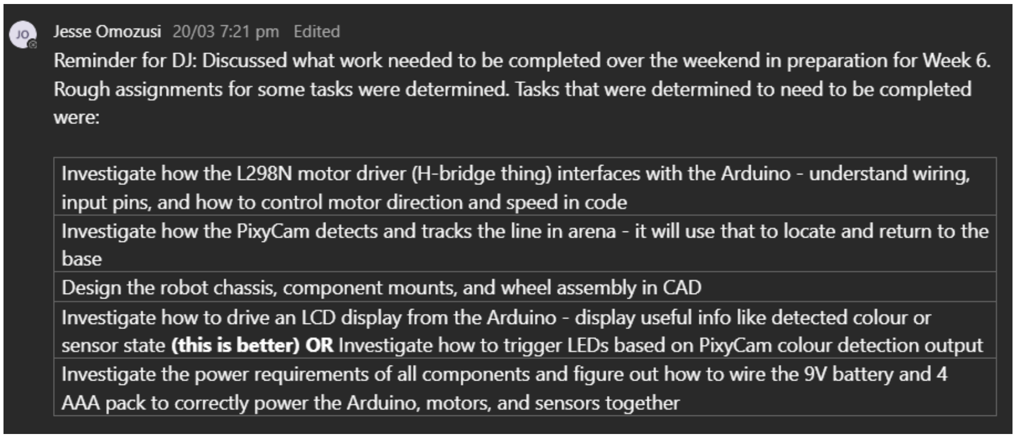
Figure 2: Flowchart evaluating S1 vs S2 feasibility for compliance testing. **Made by me**

I was leaning towards **Subsystem 1** because I had already set up and tested the PixyCam's colour signatures back in Weeks 2 and 3, so the colour detection side of things was largely sorted. On top of that, I had simulated the ultrasonic and IR sensors on TinkerCAD in Week 3, giving us a head start on obstacle detection too. Going with S2 would mean we'd need the chassis and clamp housing fully built, and the servo clamp mechanism reliable enough for 10 consecutive pickups in 90 seconds; which felt risky given where we were at. That said, the final call would be made at the subgroup leader meeting on the 19th.

In parallel, Franklin started working on the chassis design in Fusion 360 and Yusuf began looking into the power requirements and distribution for the electrical layout.

1.1.2 Entry 2 (19/03)

On the 19th of March, Jesse convened a subgroup leader meeting where we established and distributed the key tasks that need to be completed ahead of compliance testing. See **Figure** 3 for the task breakdown.



Jesse Omozusi 20/03 7:21 pm Edited

Reminder for DJ: Discussed what work needed to be completed over the weekend in preparation for Week 6. Rough assignments for some tasks were determined. Tasks that were determined to need to be completed were:

Investigate how the L298N motor driver (H-bridge thing) interfaces with the Arduino - understand wiring, input pins, and how to control motor direction and speed in code
Investigate how the PixyCam detects and tracks the line in arena - it will use that to locate and return to the base
Design the robot chassis, component mounts, and wheel assembly in CAD
Investigate how to drive an LCD display from the Arduino - display useful info like detected colour or sensor state **(this is better) OR** Investigate how to trigger LEDs based on PixyCam colour detection output
Investigate the power requirements of all components and figure out how to wire the 9V battery and 4 AAA pack to correctly power the Arduino, motors, and sensors together

Figure 3: Task breakdown established at the subgroup leader meeting. **Organised by Jesse**

After the meeting, I got straight into my assigned tasks. The first thing I tackled was the **PixyCam**. Through reading the Pixy2 documentation, I discovered that the Pixy2 **cannot run line tracking and Colour Connected Components (CCC) mode simultaneously**. This was a significant finding because it meant that for the "return to base" requirement in compliance testing, we could not rely on line following while also using the camera for colour detection. Instead, **base colour recognition** would be our approach for navigation back to the starting point.

I also looked into how the Pixy2 communicates with the Arduino Uno. The Pixy2 connects via the **ICSP header** using SPI protocol, which provides a two-way communication link. The Arduino can request block data (detected colour signatures, X-Y coordinates, signature size) from the Pixy2, and the Pixy2 receives 5V power through the same ICSP connection [1, 2]. Understanding this was important for planning how to integrate the camera data into the main control loop.

Next, I worked on the **information display** for compliance testing. Since S1 does not allow serial monitor output, I needed an alternative way to display the robot's response to detected colours. I initially tried using LEDs but they were a hassle to get working properly with the limited remaining pins. So I switched to using a **16x2 LCD display (1602)**. At first I did not have an I2C backpack module, so I wired it directly to the Arduino using individual pin connections with a potentiometer for contrast adjustment. The connection layout was as follows:

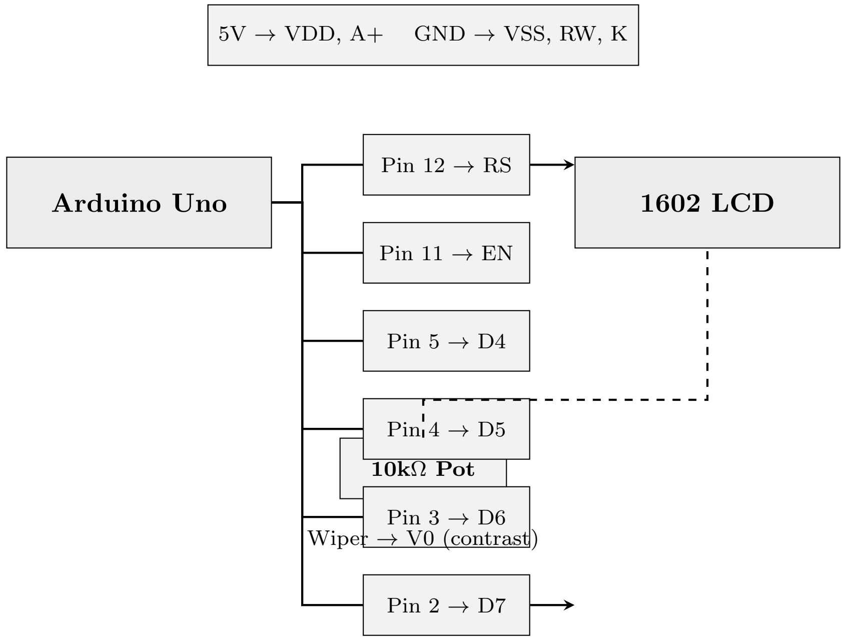


Figure 4: LCD 1602 direct wiring layout to Arduino Uno (without I2C backpack). **Made by me**

I got the LCD displaying text successfully, but realised that this direct wiring method uses up a lot of Arduino pins (6 data/control pins) which we will need for motors and sensors later. So I ordered an **I2C backpack** for the LCD which would reduce the connection to just 2 pins (SDA and SCL), freeing up the others.

Additionally, during the subgroup leader meeting, we figured out that a **separate 9V battery** is required for the MKRVRS motor driver. This is because the DC motor stall current is higher than what the Arduino's onboard power can supply, so an external source is needed to avoid voltage drops and potential damage. The power source for the servos and Arduino itself were still to be determined at this point.

1.1.3 Entry 3 (21/03)

On the 21st of March, I got the **Pixy2 camera and LCD display working together** with the Arduino. This was the first time we had a fully functional colour detection system that could identify a ball colour and display the appropriate action on screen without any laptop connection; exactly what compliance testing S1 requires.

I tested all three ball colours (red, green, blue) by placing the foam balls in front of the Pixy2 and checking whether the LCD correctly displayed the corresponding action. For red (signature 1), the LCD would display "Pick – RED (HDPE)." For green (signature 2), it displayed "Pick – GREEN (PET)." For blue (signature 3), it displayed "Pick – BLUE (PP)." And when a yellow ball was presented (or no target colour detected), the display showed an ignore/avoid response. See **Figure** 5 for the LCD displaying a detection result and **Figure** 6 for my Teams post communicating the results to the team.

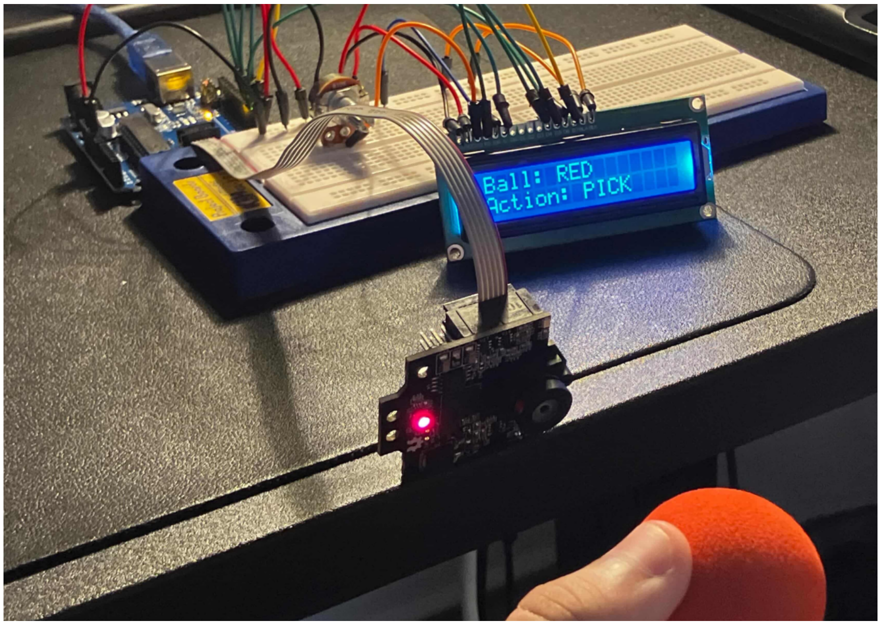


Figure 5: LCD displaying colour detection result from Pixy2 during testing. **Done by me**

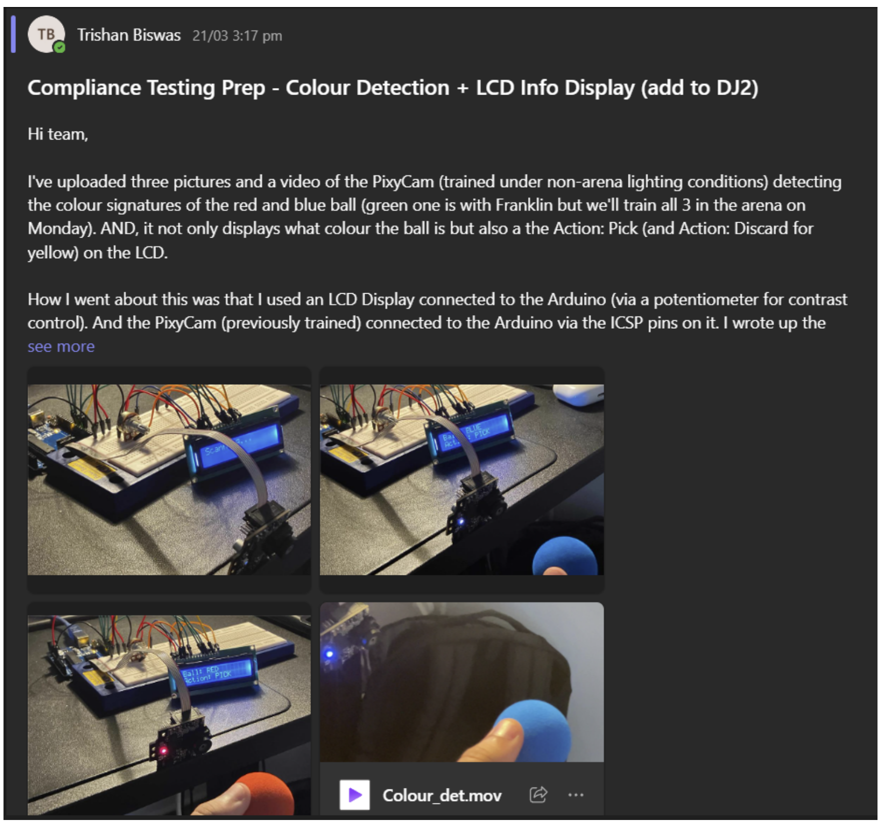


Figure 6: Teams post communicating the Pixy2 + LCD test results to the team. **Posted by me**

The main challenge I ran into was learning how to print text on the LCD via C code on the Arduino IDE. The `LiquidCrystal` library handles this, but getting the cursor positioning right for multi-line output and clearing the display between detections took some trial and error. Once I got the hang of `lcd.setCursor()` and `lcd.print()`, it was smooth from there.

This was still using the **direct-wired LCD without the I2C backpack**, and through this test it became even more clear that we need the I2C module. The LCD alone is using 6 digital pins on the Arduino, and once we add the motor driver (4 pins), ultrasonic sensor (2 pins), IR sensor (1 pin), and servos (2 pins), we simply do not have enough pins left. The I2C backpack will reduce the LCD to just 2 pins (SDA and SCL), which frees up 4 pins for other components.

I also recorded a video of the full test and shared a thumbnail on Teams for the team to see. See **Figure** 7 for the video thumbnail.

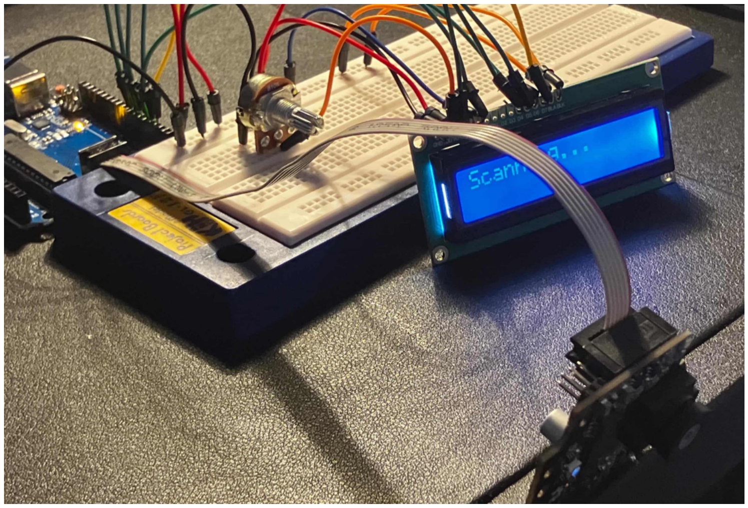


Figure 7: Thumbnail of the Pixy2 + LCD colour detection test video. **Recorded by me**

1.2 Week 6

1.2.1 Entry 4 (23/03)

On the 23rd of March, the whole team came into the ElecEng Building and Makerspace for our first proper build session from 1pm to 5pm. This was a big day as it was the first time we had all subgroups working physically on the robot at the same time.

Franklin and Bradman brought in the 3D printed wheels for testing and laser cut a test plywood chassis. They also 3D printed DC motor mounts which they screwed into the plywood to create a test chassis that we could use for integration testing. See **Figure** 8 for Franklin's wheel dimension diagram and **Figure** 9 for the test chassis setup with motors mounted.

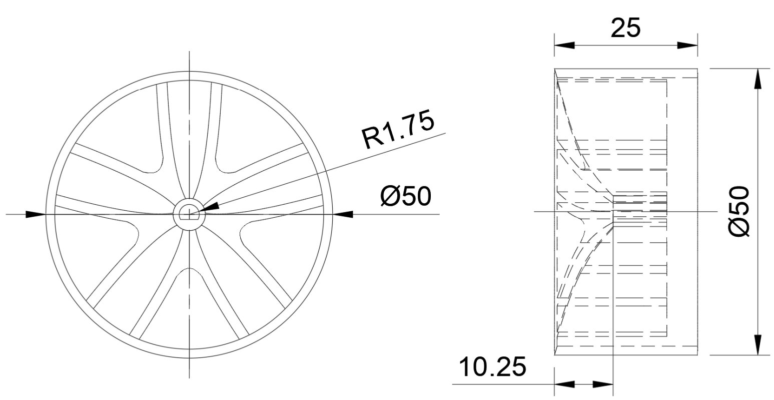


Figure 8: Wheel dimension diagram for the 3D printed wheels. **Made by Franklin**

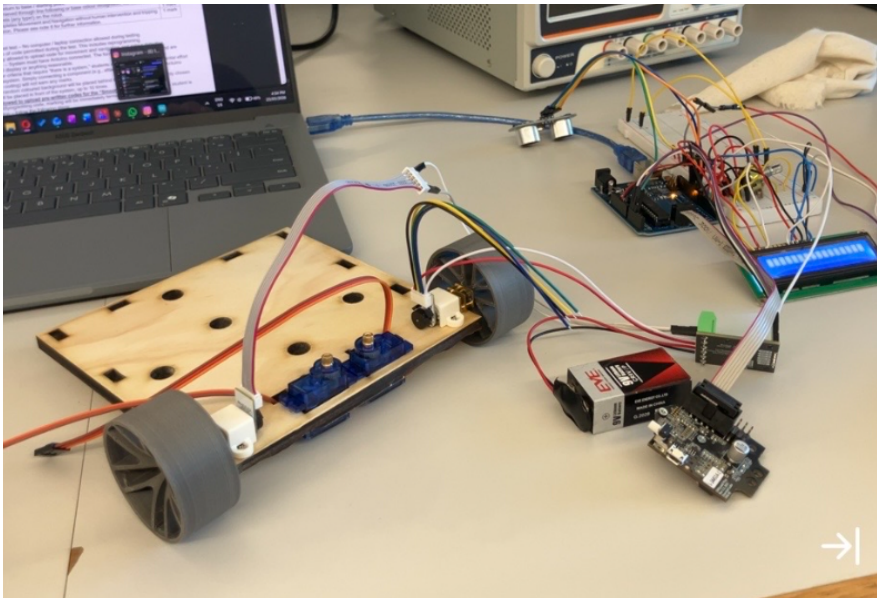


Figure 9: Test plywood chassis with DC motors and wheels mounted for integration testing. **Photo by me**

Yusuf then worked with Bradman and Franklin to set up the **MKRVRS motor driver** pinouts with the DC motors. They wrote a basic test code to run both motors and it worked first try, which was a good sign. In parallel, Jesse trained the PixyCam colour signatures more thoroughly; he tweaked the block filtering, autoexposure, and camera brightness settings to get more reliable detection under the lab lighting conditions.

Once all of that was done, **I took the test chassis to the lab** and began my part of the work. First, I tested motor movement with **PWM tuning** across different movement directions; forward, backward, and turning. I experimented with different PWM values to find a speed that was controllable but not too slow. Then, I integrated an **IR sensor** with the motor control code to set up a basic obstacle detection proof of concept. The logic was straightforward:

- Motors run forward continuously at a set PWM
- When the IR sensor detects an obstacle (outputs a LOW signal), the motors stop
- When the obstacle is removed (signal goes back to HIGH), the motors resume forward movement

This was a simpler test than using the ultrasonic sensor with a distance threshold, but it served as a quick proof of concept that the motor driver and sensor could work together through the Arduino's control logic. See **Figure** 10 for the video thumbnail of the obstacle detection test.

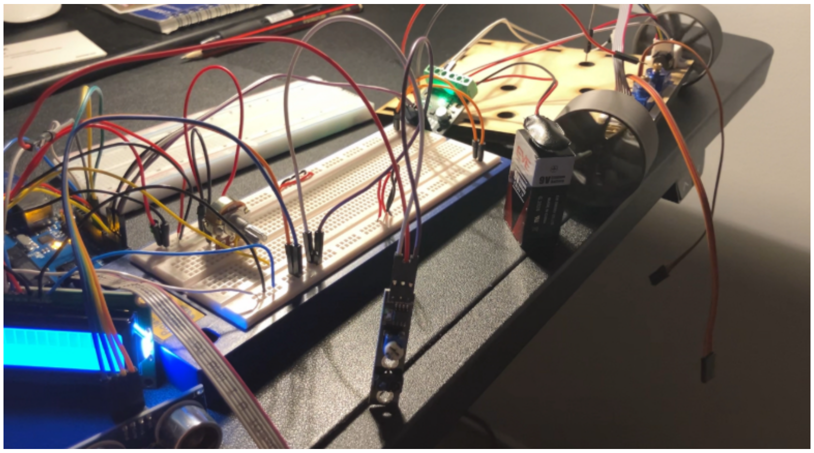


Figure 10: Thumbnail of the IR obstacle detection + motor stop/resume test. **Recorded by me**

1.2.2 Entry 5 (25/03)

On the 25th of March, the team came in again and this is when things really started coming together. Franklin and Bradman had 3D printed a proper PLA chassis plate which looked significantly better than the plywood test chassis from two days ago. They mounted the DC motors and rear caster wheel onto the chassis, and then we began mounting the **MKRVRS motor driver** on a dedicated precision mount on the chassis.

On the top plate, we mounted the **Arduino UNO mounting points and the breadboard**. Then, we attached the **PixyCam and ultrasonic sensor front plate** for our sensors. This assembly work was done by Franklin, Bradman, Jesse, and myself, with everyone present. Yusuf handled the **I2C backpack soldering** for the LCD and helped with the overall circuit pinouts. See **Figure** 11 for the assembled chassis with all components mounted.

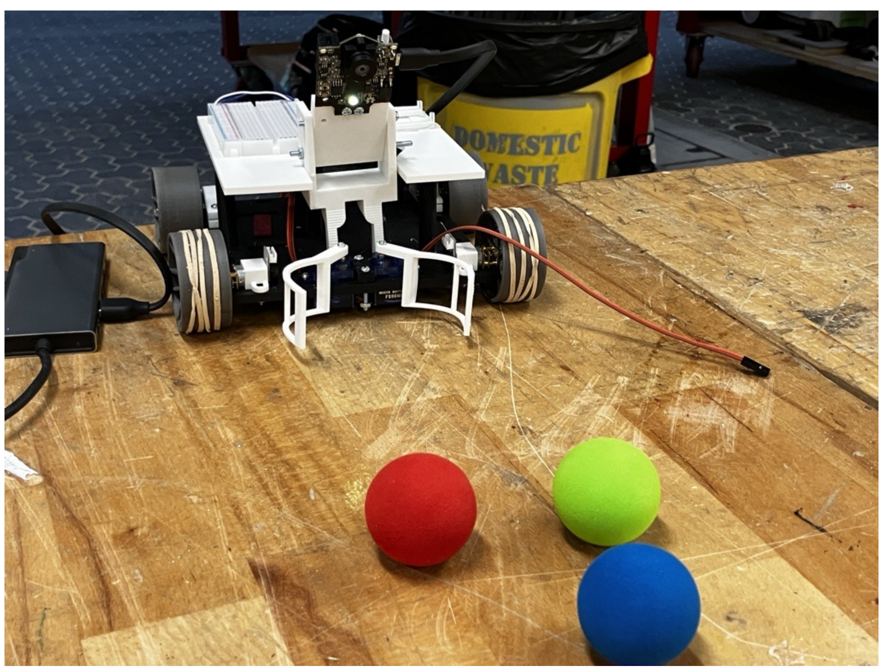


Figure 11: Assembled PLA chassis with motors, Arduino, breadboard, and sensor mounts. **Photo by me**

Once the PixyCam was mounted on the front plate, we quickly checked the **field of view (FOV)** and tuned the mounting angle so the camera could see both the ground level balls and the base colour plates. See **Figure** 12 for the PixyCam successfully detecting a colour signature at the mounted angle.

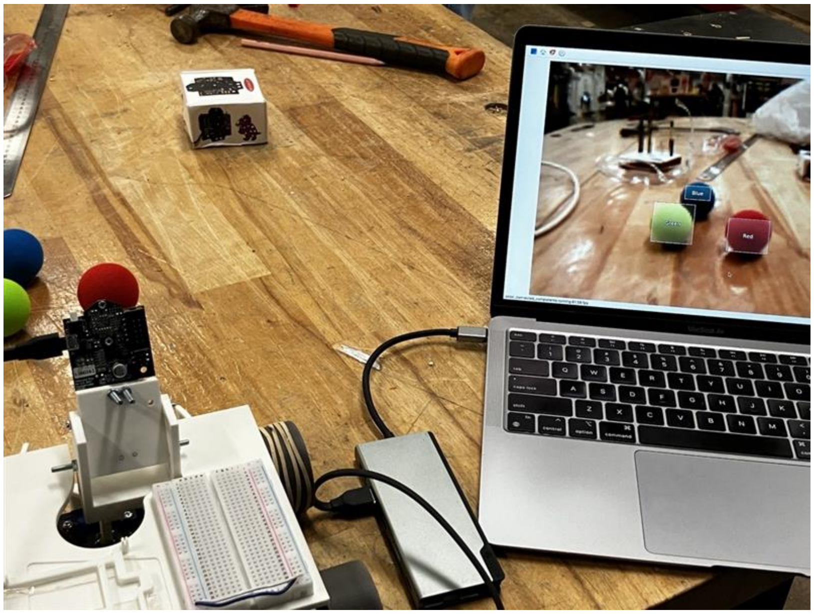


Figure 12: PixyCam mounted on the chassis and successfully detecting a colour signature. **Photo by me**

After the session, **I took the robot home** and rewrote the obstacle detection code to use the **ultrasonic sensor** instead of the IR sensor. I set a distance threshold of 20cm; when the ultrasonic reading drops below 20cm, the motors stop, and when the obstacle is cleared and the reading goes back above 20cm, the motors resume. This worked as expected and was a significant step up from the IR-based proof of concept because we now have actual distance data to work with for the compliance testing demo. See **Figure** 13 for the video thumbnail of this test.

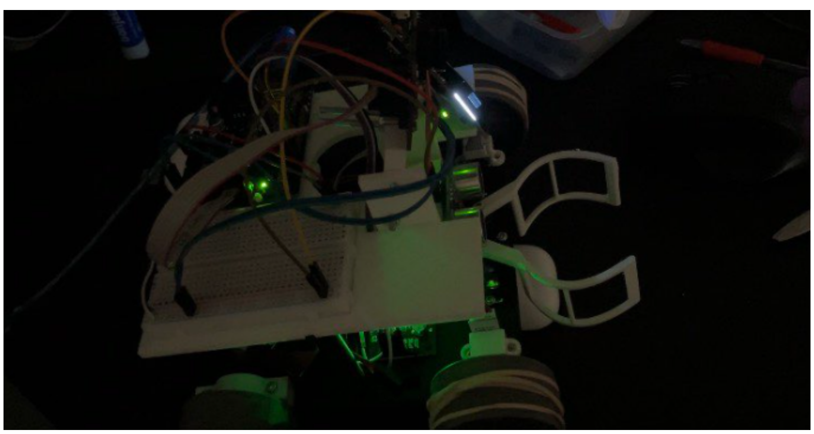


Figure 13: Thumbnail of the ultrasonic obstacle detection test (20cm threshold). **Recorded by me**

1.2.3 Entry 6 (26/03)

On the 26th of March, we fully assembled and wired the robot for compliance testing. This was the critical debugging day we had targeted since the start of Week 5. The chassis was finalised, all components were wired to the Arduino, and we connected the power supplies: a **9V lithium polymer battery** for the motors via the MKRVRS driver and a **9V battery via barrel jack** to power the Arduino. See **Figure** 14 for the robot fully assembled and ready for compliance testing.

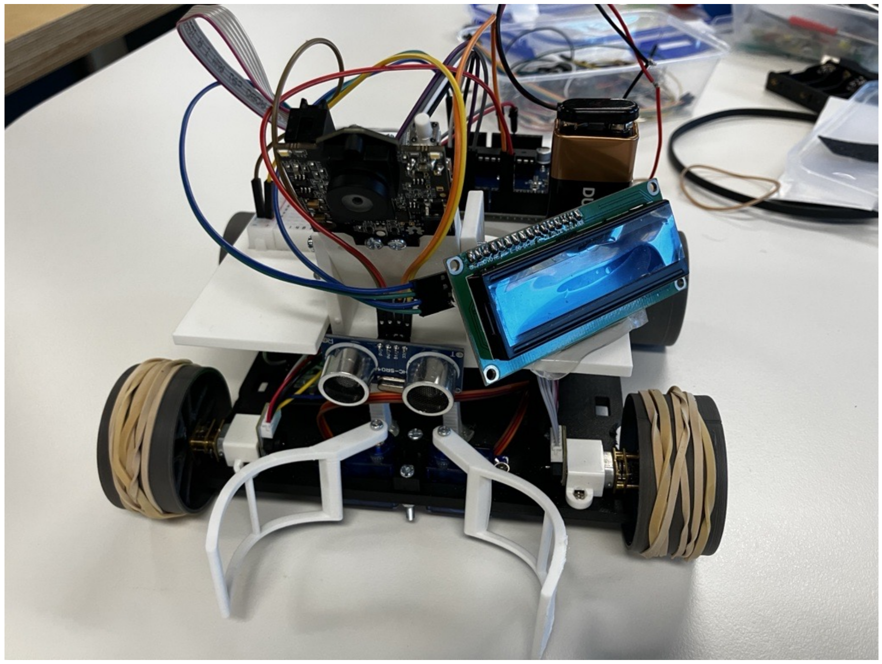


Figure 14: Robot fully assembled and wired for compliance testing. **Photo by me**

The overall circuit layout for the S1 compliance test configuration (without the IR clamp sensor and servos, as they are not needed for S1) is shown in **Figure** 15.

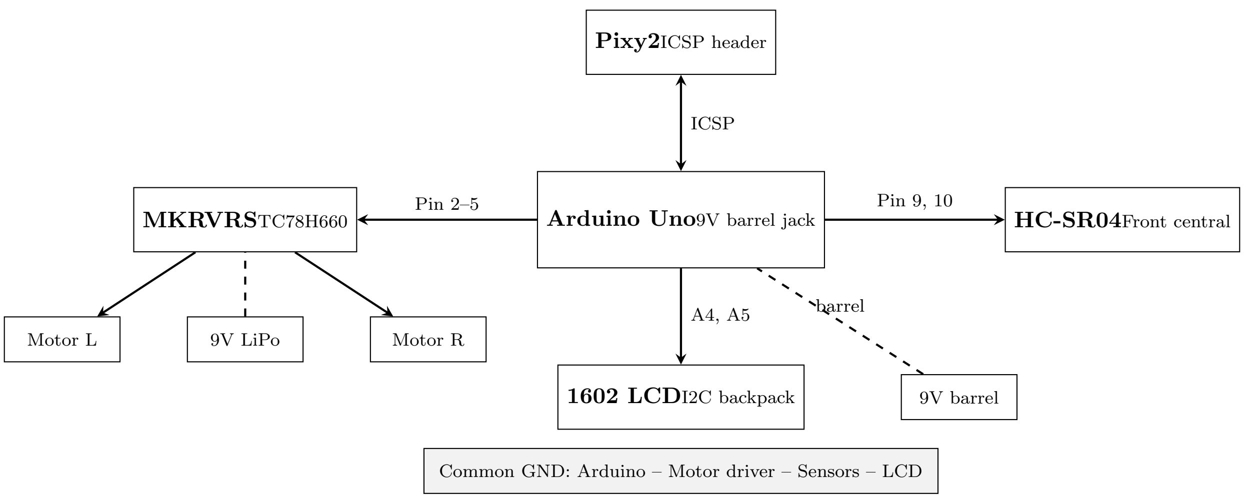


Figure 15: S1 compliance testing circuit layout (IR clamp sensor and servos excluded). **Made by me**

Once the wiring was done, we tested the **obstacle detection system** on the floor. We set the robot moving forward at 100 PWM and placed our foot in front of it; it stopped, and resumed once the foot was removed. For the ultrasonic sensor, I carried over the **median filter** from Design Journal 1 to smooth out noisy readings. However, I then realised that in the actual compliance testing arena, **multiple robots will be operating simultaneously**, and their ultrasonic bursts could trigger false readings on our sensor. To address this, **I devised a time jitter technique** that randomises the trigger timing between readings so our sensor does not synchronise with other robots' ultrasonic bursts [3]. See **Figure** 16 for the time jitter code.

```
void loop() {
    // Read distance via median filter
    int distance = medianFilter();

    if (distance > 0 && distance < OBSTACLE_THRESHOLD) {
        stopMotors();
    }

    // Random delay breaks synchronisation with other
    // robots' ultrasonic sensors to prevent crosstalk
    delay(random(50, 150));
}
```

Figure 16: Time jitter technique to prevent ultrasonic crosstalk between robots. **Coded by me**

Next, we **calibrated the rotation durations and PWM speeds** for the different movement modes. Through trial and error on the arena floor, we settled on the following values:

```
// Movement PWM values
#define FORWARD_PWM      80
#define BACKWARD_PWM     60
#define TURN_PWM         40
#define SLOW_APPROACH    30

// Rotation calibration
#define TURN_180_MS      1900   // ms for 180 degree rotation
#define TURN_360_MS      3800   // ms for 360 degree rotation
```

Figure 17: Calibrated PWM speeds and rotation durations. **Calibrated and coded by me**

After the rest of the team left, **Jesse and I stayed late until 9pm** to test the base colour recognition system. Initially, we tried using **raw RGB signature matching** to identify the base colour, but it was inconsistent. As the robot moved around the arena, the lighting changed at different positions which caused the RGB values to shift. We tested it and it only worked **2 out of 20 times**. We then switched to **CCC training the base colour signatures** directly in PixyMon; the same way we train the ball colours. After retraining, we fed the base signature data into the algorithm and it worked **every single time**. This confirmed that CCC-based base recognition is far more reliable than raw RGB matching.

With the base colour recognition sorted, I wrote the **compliance testing algorithm** that ties everything together. The flow is shown in **Figure** 18.

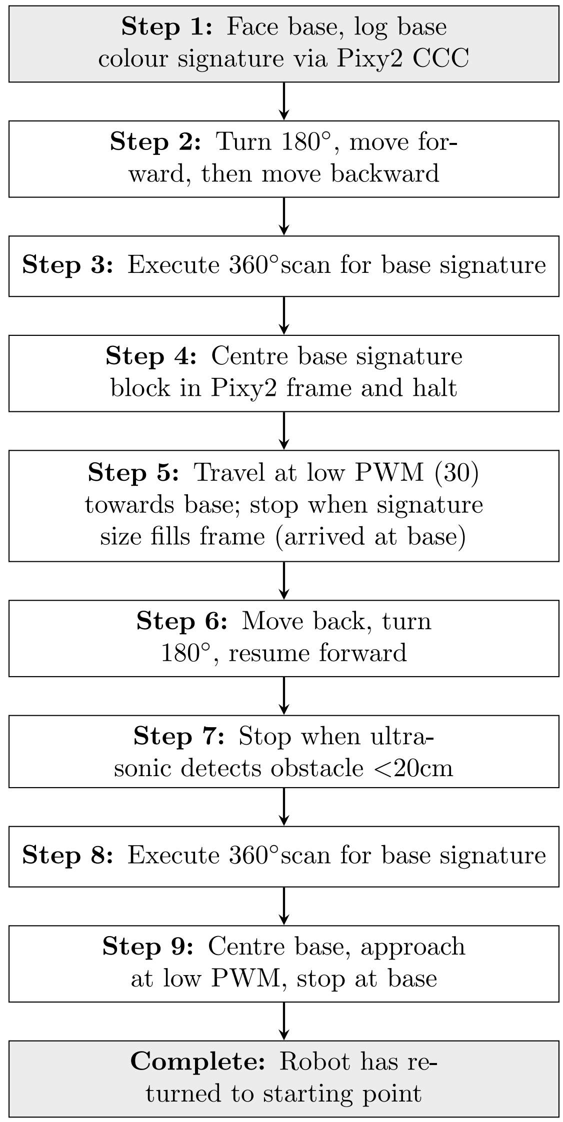


Figure 18: S1 compliance testing algorithm for navigation and base return. **Designed and coded by me**

1.2.4 Entry 7 (28/03)

On Sunday the 28th of March, the team came in for a full dry run ahead of Monday's compliance testing. This was our final chance to test the robot in the arena and iron out any remaining issues before the real thing.

We started by **calibrating the ultrasonic threshold distance** for obstacle detection. The previous 20cm threshold from Entry 5 was a starting point, but we needed to verify it worked under the actual arena conditions and CT marking criteria. We ran several tests where the robot would move forward and stop when approaching an obstacle, then resume once cleared. The obstacle detection was confirmed working at the calibrated threshold.

Next, we moved on to testing the turns. This is where we ran into a major issue. The **180° and 360° rotations were inconsistent** every single time. After watching the robot attempt several turns, we identified the problem: the **two rear wheels were dragging** during rotation. Because both rear wheels are fixed and not powered, they resist the turning motion of the front differential drive, causing the robot to pivot inconsistently depending on minor surface variations and battery voltage.

The solution was straightforward. We **removed the two rear wheels and replaced them with a single rear caster wheel**. A caster wheel can rotate freely in any direction, which means the robot can now essentially **rotate on the spot** without dragging resistance. Franklin and Bradman located the centrepoint on the rear of the chassis and bolted the caster wheel on securely. See **Figure** 19 for the robot standing on the single rear caster.

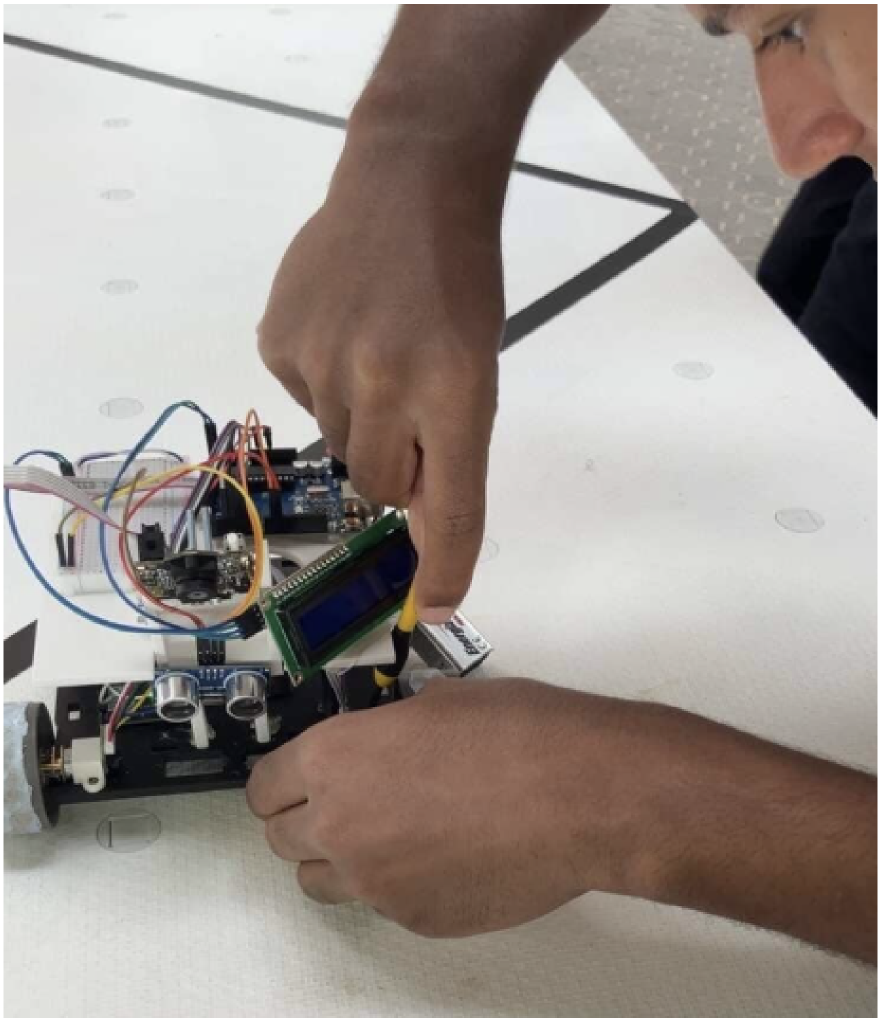


Figure 19: Robot standing on single rear caster wheel after removing the two fixed rear wheels. **Photo by me**

With the mechanical change done, **I recalibrated all the turn durations and PWM values**. The reduced friction from the caster wheel meant the robot turned faster and more consistently, so the PWM values needed to go **5–10 PWM lower** to maintain control. The new calibrated values were:

```
// Recalibrated PWM values (caster wheel)
#define FORWARD_PWM       75   // was 80
#define BACKWARD_PWM      55   // was 60
#define TURN_PWM          35   // was 40
#define SLOW_APPROACH     25   // was 30

// Recalibrated rotation durations
#define TURN_180_MS       1750 // was 1900
#define TURN_360_MS       3500 // was 3800
```

Figure 20: Recalibrated PWM speeds and rotation durations after caster wheel installation. **Calibrated and coded by me**

After the movement calibration, we tested the **base colour recognition system**. This revealed another issue. The PixyCam was detecting the **base plate's edge signature** and stopping the scan immediately, then moving towards the edge rather than the centre of the base. This would cause the robot to approach the base at an angle and potentially miss it entirely.

The correct behaviour needed to be: the robot must do a **side-to-side sweep** until the base signature block is **centred in the Pixy2's frame**, and only *then* halt the scan and begin the approach. Once approaching, the robot should stop when the signature size fills the frame, indicating it has arrived at the base. **I modified the base recognition logic** to include this centring step. See **Figure** 21 for the updated base approach algorithm.

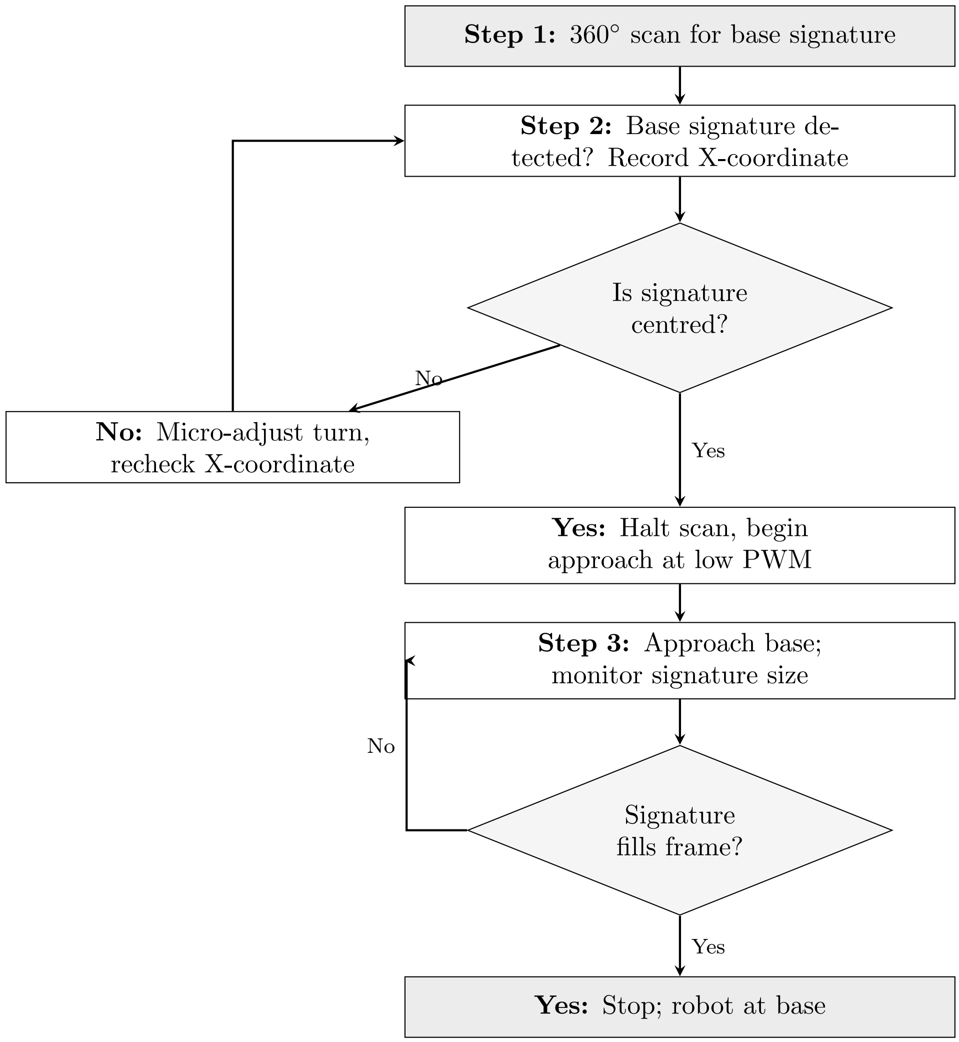


Figure 21: Updated base approach algorithm with signature centring logic. **Designed and coded by me**

We also tested the base colour logging system at startup. When the robot is first turned on and faces the base, the PixyCam logs the base colour signature so it knows what to look for during the return navigation. See **Figure** 22 for the LCD displaying the logged base colour on startup.

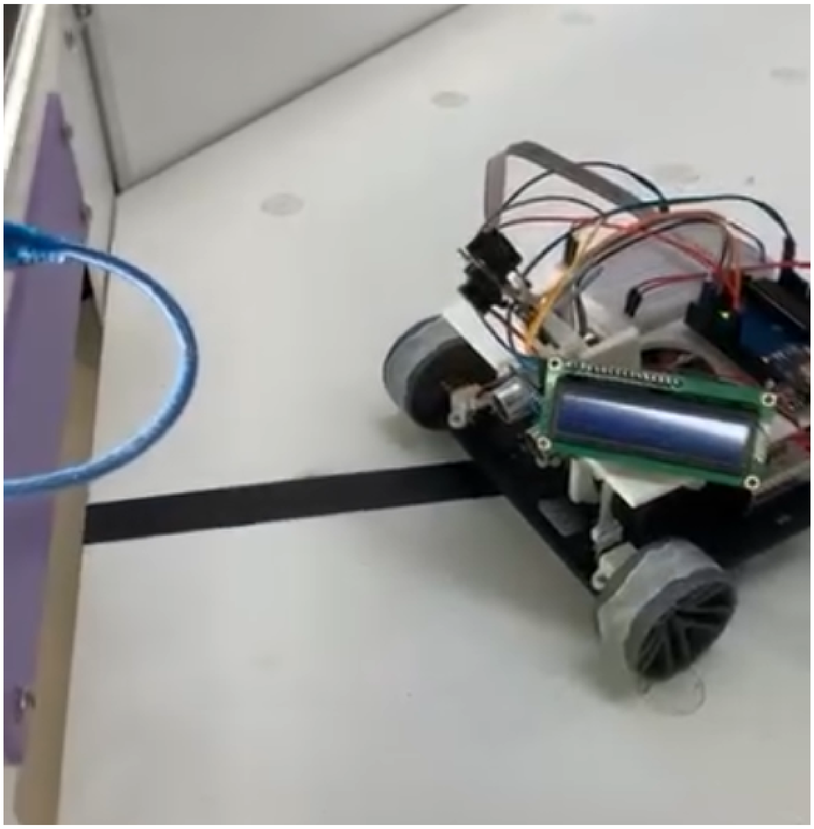


Figure 22: LCD displaying the base colour signature logged on startup. **Photo by me**

After all the fixes and recalibrations, we ran several trial runs in the arena to debug the full S1 algorithm end-to-end. See **Figure** 23 for one of the trial runs.

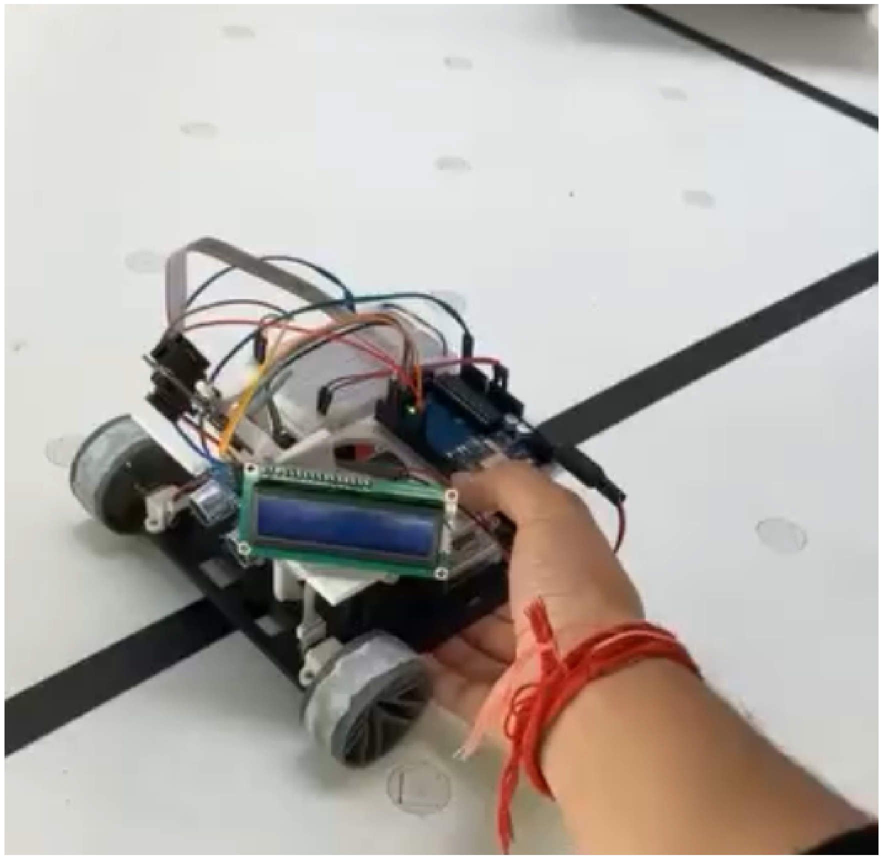


Figure 23: Trial run of the S1 algorithm in the arena during dry run debugging. **Photo by me**

By the end of the session, the mechanical issue with the rear wheels was resolved, the movement was recalibrated and consistent, and the base recognition logic had been updated. We were ready for compliance testing the next day.

1.3 Week 7

1.3.1 Entry 8 (30/03)

The 30th of March was compliance testing day. We arrived two hours early at James Kirby Makerspace, from 13:00, to run through everything before the actual CT session at 15:30. The plan was simple: electrical would handle any wiring issues, mechanical would deal with any chassis or mounting problems, and **Franklin and I handled all the coding and algorithm testing**. Jesse was there in an oversight and team leader capacity. Coming in early was essential because we needed to verify everything worked under the actual arena conditions, not just at home or in the lab.

The first thing **I focused on during the pre-CT session was locking in the RGB colour detection cycle**. The S1 compliance testing criteria requires the robot to detect a ball, display the correct response on the LCD, and cycle through the colour logic reliably. I ran the robot through repeated tests with red, green, and blue balls to make sure the PixyCam signatures were stable and the LCD output was consistent every time. A few parameter tweaks were needed to make sure no signature was being falsely triggered. See **Figure** 24 for the colour cycle detection running during pre-CT testing.

Figure 24: RGB colour cycle detection running during pre-CT testing. **Photo by me**

The second major thing I worked on was the **base colour recognition sweep**. The `centreOnBase()` function needed to reliably sweep, detect the base signature, centre it in the Pixy2 frame, and then approach. The logic works by reading the block X coordinate: if it is below 120 the robot nudges left; if it is above 196 the robot nudges right. Once the block sits within the 120 to 196 range, the robot stops adjusting and begins the slow approach, stopping when the signature size fills the frame. **I wrote and refined this function** and tested it repeatedly in the arena until it was consistently hitting the base without overshooting or drifting. See **Figure** 25 for the `centreOnBase()` function code.

```
void centreOnBase() {
  lcdMsg("Centering...", sigName(baseSig));

  for (int attempts = 0; attempts < 40; attempts++) {
    pixy.ccc.getBlocks();
    for (int i = 0; i < pixy.ccc.numBlocks; i++) {
      if (pixy.ccc.blocks[i].m_signature == baseSig) {
        int blockX = pixy.ccc.blocks[i].m_x;
        if (blockX >= 120 && blockX <= 196) {
          // Centred enough
          stopMotors();
          return;
        } else if (blockX < 120) {
          turnLeftSlow();
          delay(60);
          stopMotors();
          delay(50);
        } else {
          turnRightSlow();
          delay(60);
          stopMotors();
          delay(50);
        }
      }
    }
    delay(30);
  }
  stopMotors();
}
```

Figure 25: `centreOnBase()` function: nudges robot until base signature is centred in Pixy2 frame. **Coded by me**

The sweep-centre-approach sequence that `centreOnBase()` is part of was designed in Entry 7 and is shown in **Figure** 21. The full movement and navigation code summaries are shown in **Figures** 26 and 27.

Trishan Biswas 1/04 2:00 pm

Movement and Navigation test code summary:

Peripheral	Protocol	Purpose
Pixy2 Camera	SPI (pins 11-13)	Colour signature detection and tracking
16x2 LCD Display	I2C (address 0x27, SDA=A4, SCL=A5)	Real-time status feedback
HC-SR04 Ultrasonic Sensor	Digital GPIO (Trig=9, Echo=10)	Obstacle detection
Two DC Motors	PWM (Dir+PWM on pins 2/3 and 4/5)	Differential drive movement

Colour Signature Tracking

The Pixy2 camera operates in CCC (**Colour Connected Components**) **mode**, where colours are pre-trained on the camera and stored as **signature IDs** (integers 1-7). The robot uses signatures 4 **through** 7, mapped as:

Signature ID	Colour
4	Orange
5	Yellow
6	Pink
7	Purple

At startup, the robot reads its base signature, whichever of these four signatures the Pixy2 detects when facing the starting base. This signature is stored as an integer and used for all subsequent tracking. The Pixy2 returns detected blocks with properties including m_signature (the colour ID), m_x (horizontal position in a 316px-wide frame), m_width and m_height. The robot uses m_width * m_height (block area in pixels) as a proximity metric so basically the larger the detected block, the closer the robot is to the base and thus the robot will stop as per the code.

Figure 26: Movement and navigation code algorithm summary (part 1). **Made by me**

Configurable Parameters

Parameter	Value	Purpose
NORMAL_SPEED	80 (PWM 0-255)	Standard driving speed
APPROACH_SPEED	65 (PWM 0-255)	Reduced speed for precision manoeuvres
TURN_SPEED	80 (PWM 0-255)	Rotation speed
TURN_180_MS	1400ms	Duration motors run to achieve a 180° turn
FORWARD_MS	1500ms	Duration of initial forward movement
BACKWARD_MS	1500ms	Duration of backward movement
BASE_AREA_THRESHOLD	15000 (px)	Block area at which robot considers itself "at base"
STOP_DISTANCE_CM	20cm	Ultrasonic distance threshold for obstacle halt

1. Base identification: Pixy2 scans for a colour signature (sigs 4-7), up to 20 attempts. The detected signature becomes the homing target for the rest of the run.
2. Turn 180°: differential drive (one motor forward, one backward) for 1400ms.
3. Move forward 1500ms at normal speed.
4. Move backward 1500ms at normal speed.
5. 360° scan, rotates in 100ms slices, polling the Pixy2 before each turn slice. Up to 3 full rotations maximum to find the base signature.
6. Approach base: it drives forward at approach speed, continuously monitoring block area until it exceeds 15000px.
7. Back away 900ms from the base.
8. Turn 180° away from base.
9. Obstacle detection: SPI and I2C buses are shut down to isolate the ultrasonic sensor from interference. The robot drives forward using a 3 sample median filtered ultrasonic reading that I made using a simple sort, it essentially stops when an object is detected within 20cm.
Peripherals are then re started so the Pixy and the I2C bus.
10. Second base scan: rot it ates slowly (at approach speed, so slower) in 100ms slices. Upon any detection of the base signature, the robot stops and runs a centering routine that nudges left/right until the block's X-coordinate falls within the centre zone (120-196 of the 316px frame). It then approaches at reduced speed until block area exceeds the threshold and then stops and ceases all movement

Figure 27: Movement and navigation code algorithm summary (part 2). **Made by me**

Once both the colour detection cycle and the base navigation were confirmed working, we ran the **full movement and navigation sequence end-to-end** to verify the whole CT demo would flow without issues. See **Figure** 28 for the robot running through the full sequence during the pre-CT session.

Figure 28: Robot performing the full movement and navigation sequence during pre-CT testing. **Photo by me**

By the time we finished the pre-CT session we were confident every criteria point was covered. We headed into the actual compliance testing session at 15:30 with the robot fully calibrated and the algorithm locked.

1.3.2 Entry 9 (02/04)

On the 2nd of April, the team convened in the CompLabs from 12pm to 3pm for a Design Proposal progress check. The submission was due Sunday, so this was the opportunity to see what everyone had written and identify what needed fixing before the deadline.

I had been working on my assigned sections since Monday. For the **Relevant Knowledge** section, I wrote several subsections covering the PixyCam integration, ultrasonic sensor operation, and motor control logic. The goal of this section was to explain the technical principles behind each component choice and how they work together in our system. For example, the PixyCam subsection explained CCC mode, why we chose it over line tracking, how the ICSP header SPI connection works, and what block data (X coordinate, signature, area) the Arduino receives and acts on. The ultrasonic subsection explained the median filter and time jitter technique we developed, with the reasoning behind each. See **Figure** 29 for one of the code explanation snippets I wrote.

3.2 Vision and Detection Subsystem

3.2.1 Pixy2 Camera Operation

The Pixy2 detects coloured objects using trained colour signatures in Colour Connected Components mode [4]. Signatures 1 to 3 identify red, green, and blue balls respectively, while signatures 4 to 7 identify the four corner bases. The camera stores base colour signatures at startup. When an object is detected, the Pixy2 reports its position as X–Y coordinates and signature pixel size [5]. The camera communicates with the Arduino via SPI protocol through the ICSP pins [6], where the Arduino requests or receives data. The Pixy2 receives 5V power through the ICSP pins.

3.2.2 HC-SR04 Ultrasonic Sensor Operation

The HC-SR04 measures distance using 40 kHz ultrasonic sound waves [7]. The Arduino sends a trigger signal to the sensor, which causes it to emit a short ultrasonic burst. This sound wave travels forward until it hits an obstacle and reflects back. The sensor detects the returning echo and measures how long the pulse was absent. The Arduino then calculates the distance using this time measurement. The sensor receives 5V power from the Arduino's 5V output pin.

The ultrasonic sensor can output inaccurate readings due to external noise, which creates spikes in distance measurements. To address this, a median filter is applied in the code to discard outliers from batches of five measurements and output the median value. Figure 5 shows the median filter implementation.

```
int medianFilter() {
    int readings[5];

    // Take 5 raw readings with short gap between each
    for (int i = 0; i < 5; i++) {
        readings[i] = sonar.ping_cm();
        delay(10);
    }

    // Bubble sort readings ascending
    for (int i = 0; i < 4; i++) {
        for (int j = 0; j < 4 - i; j++) {
            if (readings[j] > readings[j + 1]) {
                int temp = readings[j];
                readings[j] = readings[j + 1];
                readings[j + 1] = temp;
            }
        }
    }

    return readings[2]; // middle value = median
}
```

Figure 5: Code excerpt of median filter implementation using bubble sort

Figure 29: Snippet from Relevant Knowledge section explaining PixyCam and ultrasonic sensor integration. **Written by me**

I also made a **TikZ circuit diagram** for the Design Proposal documenting the full Arduino pinout and all component connections. This was important because the Design Proposal needed to show the marker that our electrical design is deliberate and well-documented, not just wires plugged in wherever they fit. The diagram shows the Pixy2 on the ICSP header, the MKRVRS motor driver on pins 2–5, the HC-SR04 on pins 9 and 10, the LCD via I2C on A4 and A5, and the two power rails (9V LiPo for motors, 9V barrel jack for Arduino) with common GND. See **Figure** 30 for the circuit diagram.

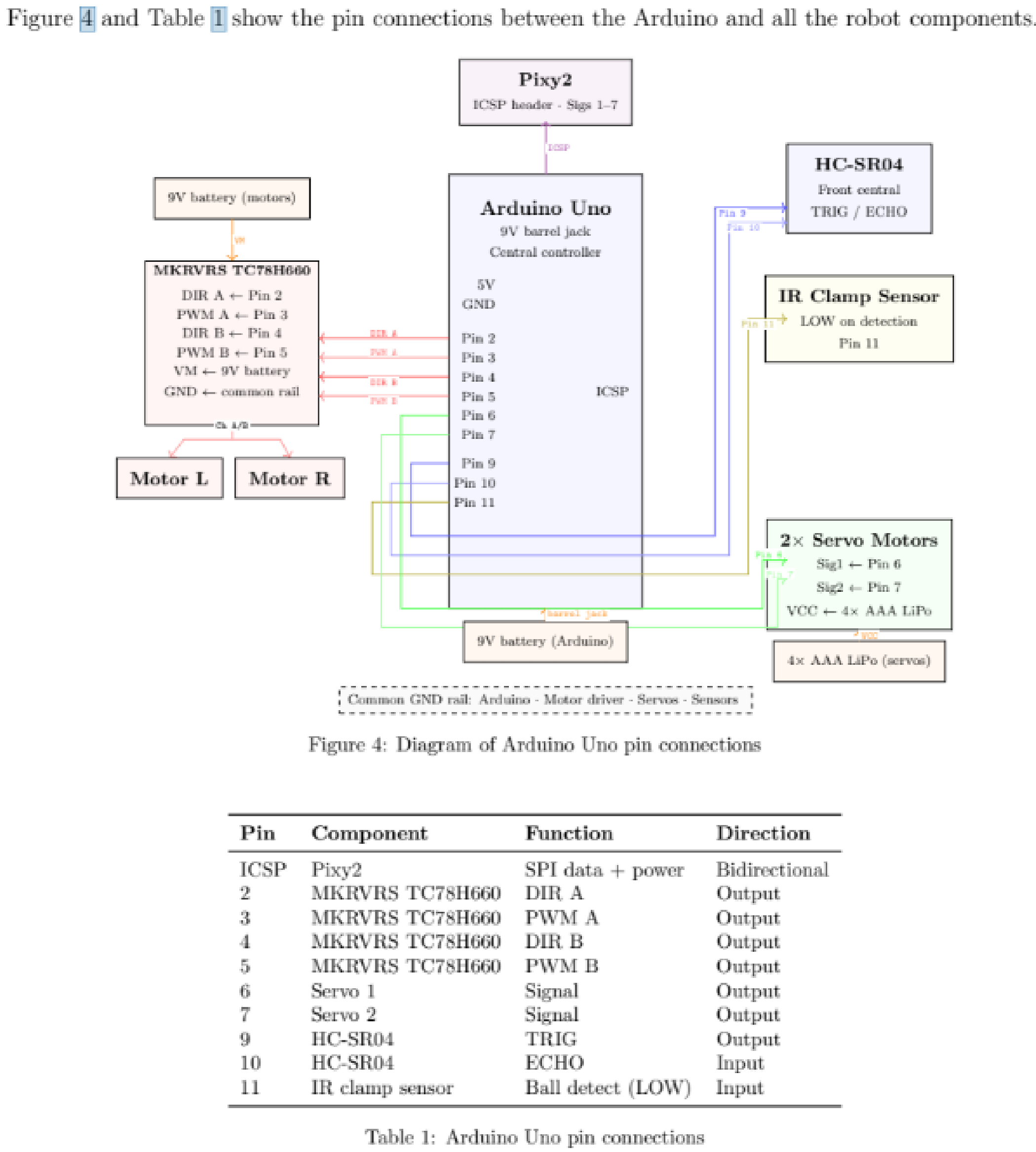


Figure 4 and Table 1 show the pin connections between the Arduino and all the robot components.

Figure 4: Diagram of Arduino Uno pin connections

Pin	Component	Function	Direction
ICSP	Pixy2	SPI data + power	Bidirectional
2	MKRVRS TC78H660	DIR A	Output
3	MKRVRS TC78H660	PWM A	Output
4	MKRVRS TC78H660	DIR B	Output
5	MKRVRS TC78H660	PWM B	Output
6	Servo 1	Signal	Output
7	Servo 2	Signal	Output
9	HC-SR04	TRIG	Output
10	HC-SR04	ECHO	Input
11	IR clamp sensor	Ball detect (LOW)	Input

Table 1: Arduino Uno pin connections

Figure 30: Arduino Uno circuit diagram with pinouts and component connections for the Design Proposal. **Made by me**

Additionally, **Jesse and I brainstormed the final testing algorithm** together. The CT demo only covered S1 (perception and navigation), but the final demonstration requires the full robot operation including ball pickup and sorting. We mapped out the high-level sequence: log base colour on startup, scan for target ball, navigate to it, collect it with the clamp, scan for the correct base, navigate to it, deposit the ball, and repeat for the R→G→B cycle. This was the first time we had the full end-to-end algorithm on paper, and having it as a flowchart gave us a clear reference for what the final testing code needed to implement. See **Figure** 31 for the final testing algorithm flowchart.

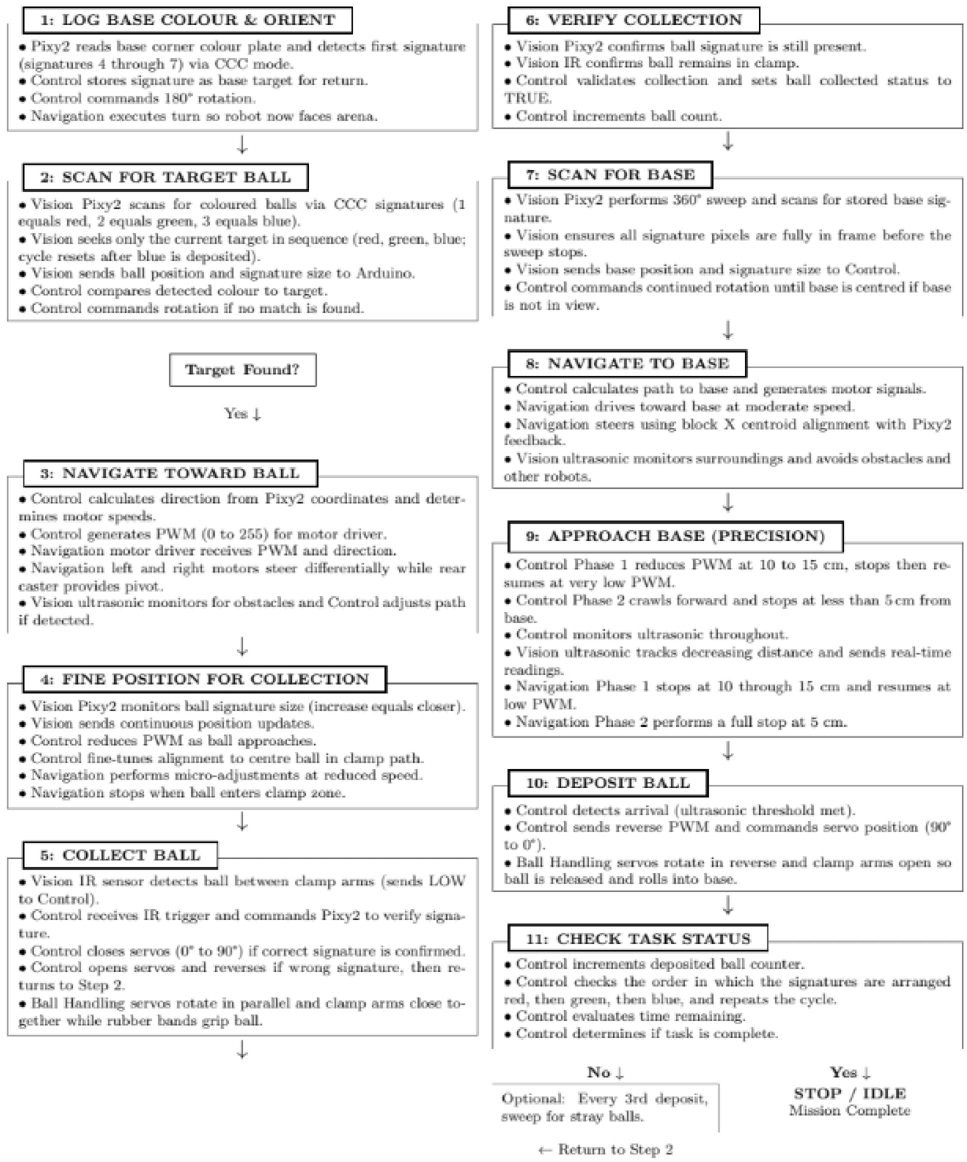


Figure 31: Final testing algorithm flowchart conceptualised with Jesse. **Made by me**

During the meeting, everyone presented their sections. Most were drafted but **needed significant refinement**; some sections were too vague and others were missing figures or technical justification. Jesse and I decided we would handle the bulk of the rewriting and editing on Saturday so that by Sunday's submission the document would be ready, and the rest of the team could shift focus to final testing preparation. **Jesse and I set up a shared Overleaf document** so we could edit simultaneously without version conflicts. Yusuf had done the majority of the bibliography and sourcing work, which saved us a lot of time on the referencing side.

1.3.3 Entry 10 (04/04)

On Saturday the 4th of April, Jesse, Yusuf, and I met virtually on Microsoft Teams to edit and format the Design Proposal. The sessions were scattered throughout the day with check-ins between independent work periods, and the main meeting ran from 5pm to 7pm. Most of the actual editing happened asynchronously on the shared Overleaf document, which made it easy to see each other's changes in real time without overwriting anything.

The main task was **transferring Word drafts into LaTeX** and making sure figures were embedded correctly. Several sections had been written in Word by team members who do not use LaTeX, so I handled the conversion: fixing formatting, adding `\noindent` and paragraph spacing, and making sure any tables or lists were properly structured. Jesse rewrote and tightened the weaker sections on his end while I focused on figure creation and layout. I also made sure the **Gantt chart** reflected our actual current progress rather than the original plan, since several tasks had shifted in timing after CT.

For figures, I created the TikZ block diagram for the system architecture (based on Jesse's sketch), formatted Franklin's wall-following diagram, and embedded the bibliography page. Having consistent TikZ figures throughout the document made it look more cohesive than a mix of screenshots and hand-drawn images. See **Figure** 32 for the wall-following figure diagram, **Figure** 33 for the updated Gantt chart, **Figure** 34 for the system block diagram, and **Figure** 35 for the bibliography.

2 Design Concept

2.1 System Overview

The proposed design is an autonomous, Wall-E-inspired robot that sorts coloured balls representing different types of plastic waste into their correct categories. The overall design of the robot is shown in Figure 1.

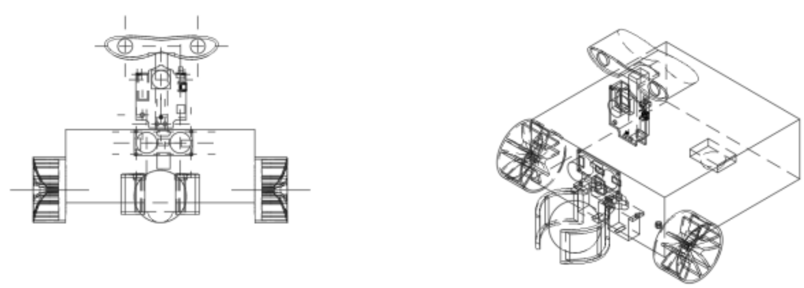


Figure 1: Technical drawing of the overall robot design

1

Figure 32: Wall-following figure diagram for the Design Proposal. **Diagram by Franklin**

5.2 Gantt Chart

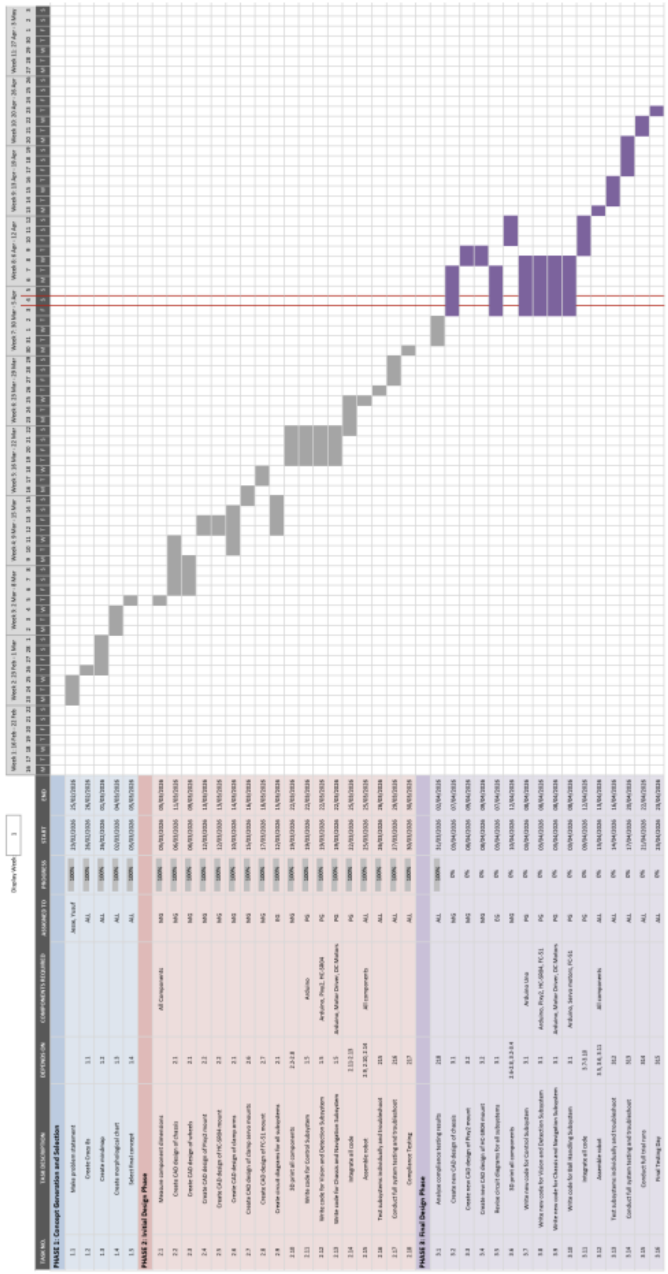


Figure 33: Gantt chart for the Design Proposal showing project timeline and progress. **Made by Jesse**

To achieve the project objectives, the robot has been divided into four distinct subsystems. This modular approach enables independent development of each subsystem and allows modifications without completely redesigning the robot. The four subsystems are:

- Control Subsystem
- Vision and Detection Subsystem
- Chassis and Navigation Subsystem
- Ball Handling Subsystem

Figure 2 illustrates the basic interactions between subsystems. Further detailed explanations of these interactions are provided throughout the Relevant Knowledge section.

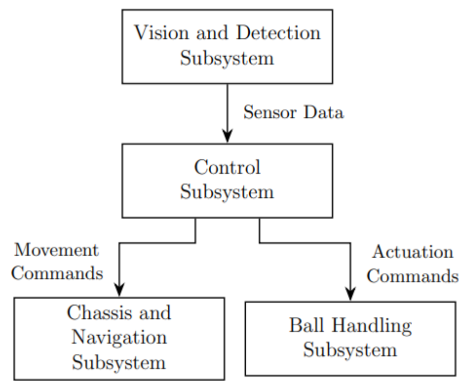


Figure 2: Block diagram of the subsystems and their interactions

2.2 Subsystem Specifications

2.2.1 Control Subsystem

Figure 34: System block diagram for the Design Proposal. **Made by Jesse**

References

[1] D. Mesta, "AI and Robotics in Waste Sorting: Enhancing Recycling Efficiency," *Advances in Recycling & Waste Management*, vol. 9, no. 4, Aug. 2024, [Online]. Available: https://www.hilarispublisher.com/open-access/ai-and-robotics-in-waste-sorting-enhancing-recycling-efficiency-109193.html. Accessed: Apr. 5, 2026.

[2] Arduino, *Arduino UNO R3 Datasheet*, rev. 3 ed., Arduino, Apr. 2024, [Online]. Available: https://docs.arduino.cc/resources/datasheets/A000066-datasheet.pdf. Accessed: Apr. 5, 2026.

[3] ——, "Secrets of Arduino PWM," [Online], May 2024, available: https://www.arduino.cc/en/Tutorial/SecretsOfArduinoPWM. Accessed: Apr. 5, 2026.

[4] Pixycam, "Color Connected Components," [Online], Oct. 2018, available: https://docs.pixycam.com/wiki/doku.php?id=wiki:v2:color_connected_components. Accessed: Apr. 5, 2026.

[5] ——, "CCC API," [Online], Oct. 2018, available: https://docs.pixycam.com/wiki/doku.php?id=wiki:v2:ccc_api. Accessed: Apr. 5, 2026.

[6] Arduino, "SPI," [Online], Jan. 2026, available: https://docs.arduino.cc/language-reference/en/functions/communication/SPI/. Accessed: Apr. 5, 2026.

[7] A. Prabhu, "How HC-SR04 Ultrasonic Sensor Works & How to Interface It With Arduino," [Online], Jan. 2026, last Minute Engineers. Available: https://lastminuteengineers.com/arduino-sr04-ultrasonic-sensor-tutorial/. Accessed: Apr. 5, 2026.

[8] S. Farah, D. G. Anderson, and R. Langer, "Physical and mechanical properties of PLA, and their functions in widespread applications—A comprehensive review," *Advanced Drug Delivery Reviews*, vol. 107, pp. 367–392, Dec. 2016.

Figure 35: Bibliography compilation for the Design Proposal. **Done by me and Yusuf**

Yusuf handled the bibliography formatting in BibTeX, and we cross-checked citations against the sources to make sure everything was accurately attributed. By Sunday morning the document was fully formatted and submission-ready. We ran a final pass for spelling, broken references, and figure numbering before submitting.

1.4 Week 8

1.4.1 Entry 11 (06/04)

On the 6th of April, the full team met in Room 412 of the Main Library from 2pm to 3pm. This was a short meeting to do a progress check and read through the final testing notification together before the push towards final demonstration day.

We read through the **Final Testing Specification** as a team and made notes on the marking criteria. I made a quick block chart evaluating the requirements and what we need to hit for an HD. See **Figure** 1.4.1 for my evaluation.
We established action items to complete by Thursday's meeting (9th April). I ordered a replacement LCD for $5 and tested it with a basic HelloWorld print. See **Figure** 36.

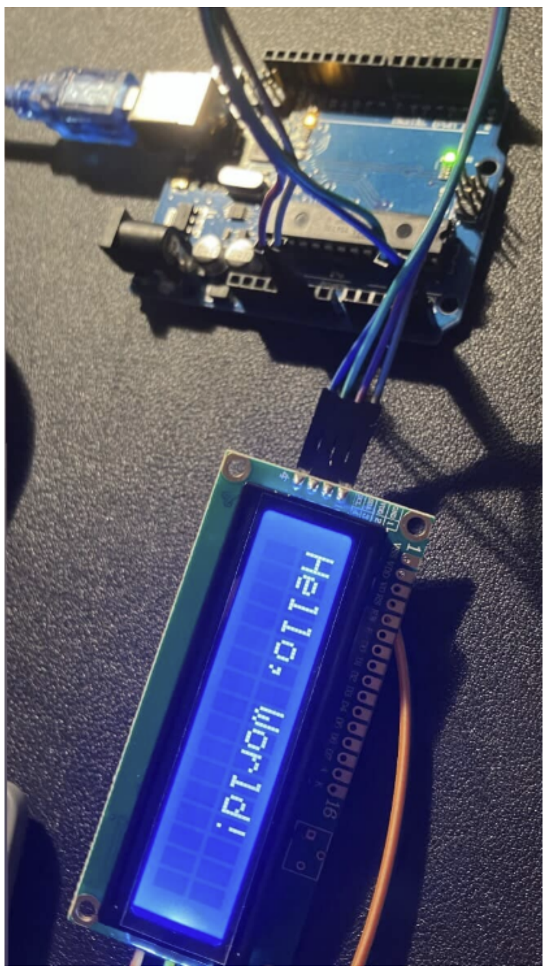


Figure 36: Replacement LCD tested with HelloWorld print. **Photo by me**

Franklin also made a diagram showing where the LCD would mount on a skeleton structure on top of the robot body. See **Figure** 37.

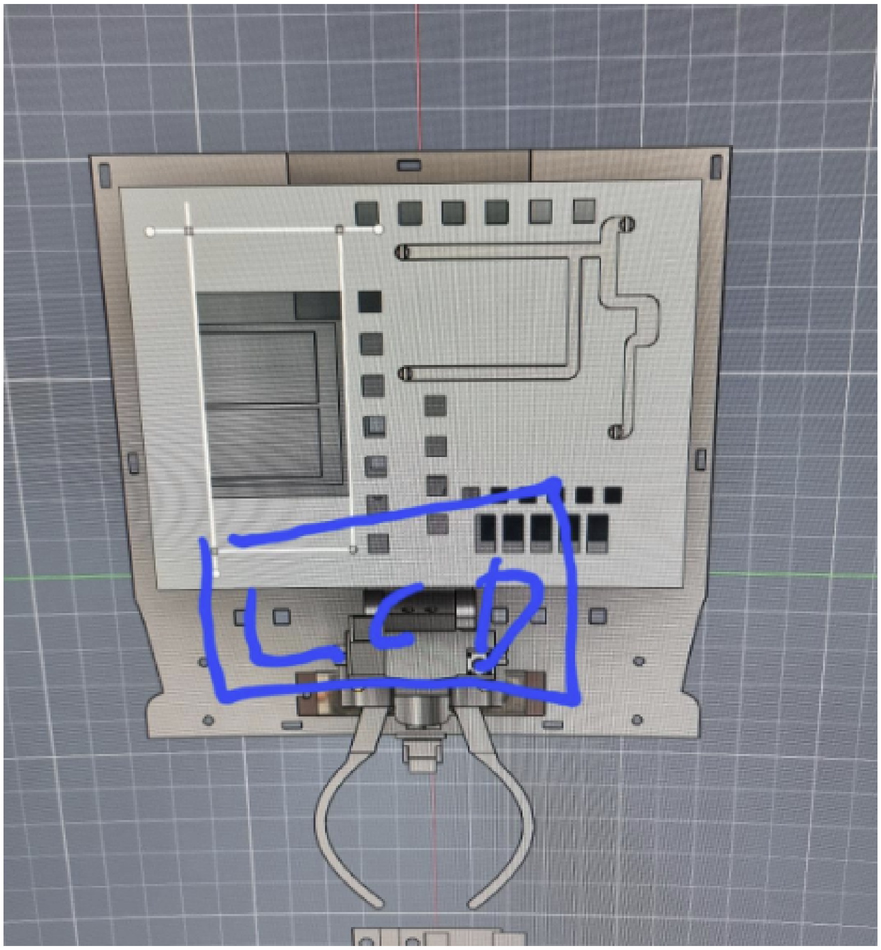


Figure 37: LCD mount placement diagram for the robot. **Made by Franklin**

We also determined the **optimal PixyCam angle and height**: 10cm from the top of the Pixy, angled at 20–30 degrees downwards for best ball detection. Franklin and Bradman began figuring out a 3D printed mould solution with silicone curing for wheel covers (instead of blu tac). Roles were assigned: Franklin to design mechanical components (LCD mount, upper level with perfboard and Arduino, bottom level with motor driver and servo batteries), Yusuf to solder everything directly without header pins, Bradman to assist Franklin, and Jesse to oversee and assist where needed.

1.4.2 Entry 12 (09/04)

On the 9th of April, the team met to follow up on the action items from Monday. We confirmed the **optimal PixyCam mounting angle** and tested the positioning. Franklin and Bradman **3D printed the wheel mould** for silicone curing and printed the **LCD mount**; we were waiting for the next print cycle.

I completed the **final testing code** (final_testing.ino) which implements the full ball pickup and sorting algorithm for HD target. The code follows a strict R→G→B cycle and handles all 11 steps of the operation. See **Figure** 38 for the configuration section and **Figure** 39 for the main run sequence.

```
namespace Config {
  // Motor pins
  constexpr int DIR_A = 2;
  constexpr int PWM_A = 3;
  constexpr int DIR_B = 4;
  constexpr int PWM_B = 5;

  // Speeds (PWM 0-255)
  constexpr int NORMAL_SPEED = 80;
  constexpr int APPROACH_SPEED = 65;
  constexpr int CRAWL_SPEED = 40;
  constexpr int TURN_SPEED = 80;

  // Timing
  constexpr int TURN_180_MS = 1400;

  // Thresholds
  constexpr int BALL_AREA_THRESHOLD = 8000;
  constexpr int BASE_AREA_THRESHOLD = 15000;
  constexpr int OBSTACLE_DISTANCE_CM = 20;

  // Ball signatures (Pixy2 CCC)
  constexpr int SIG_RED = 1;
  constexpr int SIG_GREEN = 2;
  constexpr int SIG_BLUE = 3;

  // Base signatures
  constexpr int SIG_ORANGE = 4;
  constexpr int SIG_YELLOW = 5;
  constexpr int SIG_PINK = 6;
  constexpr int SIG_PURPLE = 7;
}
```

Figure 38: Configuration namespace from final_testing.ino: pin definitions, speeds, thresholds, signatures. **Coded by me**

```
void runSequence() {
  step1_logBaseAndOrient();   // Log base colour, turn 180

  while (true) {
    step2_scanForTargetBall(); // 360 scan for R/G/B
    step3_navigateToBall();    // Centre and approach
    step4_finePosition();      // IR fine positioning
    step5_collectBall();       // Close clamp if correct
    step6_verifyCollection();  // Confirm ball secured
    step7_scanForBase();       // 360 scan for base
    step8_navigateToBase();    // Centre and approach base
    step9_approachBase();      // Ultrasonic precision
    step10_depositBall();      // Open clamp, back away

    if (step11_checkStatus()) break;
    // Advance: R->G->B cycle
  }
  lcdShow("Mission", "Complete!");
}
```

Figure 39: Main run sequence: 11-step cycle for ball collection and base deposit. **Coded by me**

The meeting also established the **Week 9 timeline and deadlines**:

Mon 14 Apr	**Tue-Wed 15-16**	**Thu 17 Apr**	**Fri-Sun 18-20**
Programming: code mostly done	3D print robot	Train Pixy2	Test & troubleshoot
Mechanical: chassis CAD	Assemble chassis & circuitry	Practice runs	Paint finish
Electrical: circuit diagram	Code deadline	Troubleshoot	Final checks

Figure 40: Week 9 timeline and deadlines established at the meeting. **Made by me**

1.5 Week 9

1.5.1 Entry 13 (13/04)

On the 13th of April, the team met in Room 412 of the Main Library. This was later than planned in the Week 9 timeline; we should have been better with time management. The meeting focused on briefing everyone on assembly and code before the build.

I presented the overall algorithm to the team in detail, walking through each step and explaining how the final_testing.ino draft code implements it. The 11-step sequence (log base, scan for ball, navigate, collect, scan for base, deposit, repeat) was now clear to everyone. Yusuf explained how he would solder all electronic components to the perfboard. Franklin showed the CAD models he had sent for printing: the clamps, base plate, top plate, LCD mount, and a casing that covers the inner skeleton to make the robot look like Wall-E.

After the briefing, the three subgroup leaders (Jesse, Franklin, Yusuf) brainstormed to ensure everything was in order: pinouts routing through the LCD mount to the Arduino shield, servo bracket placement avoiding conflict with ultrasonic and IR sensor layout, and general wire management.

We waited for the print jobs to complete. Once ready, **Franklin and Yusuf assembled the chassis** and placed components on their precision mounts: motor driver, PixyCam, IR sensor, ultrasonic, Arduino board, perfboard, and battery casings. I mounted the servos onto the bracket. See **Figures** 41, 42, and 43 for the assembly progress.

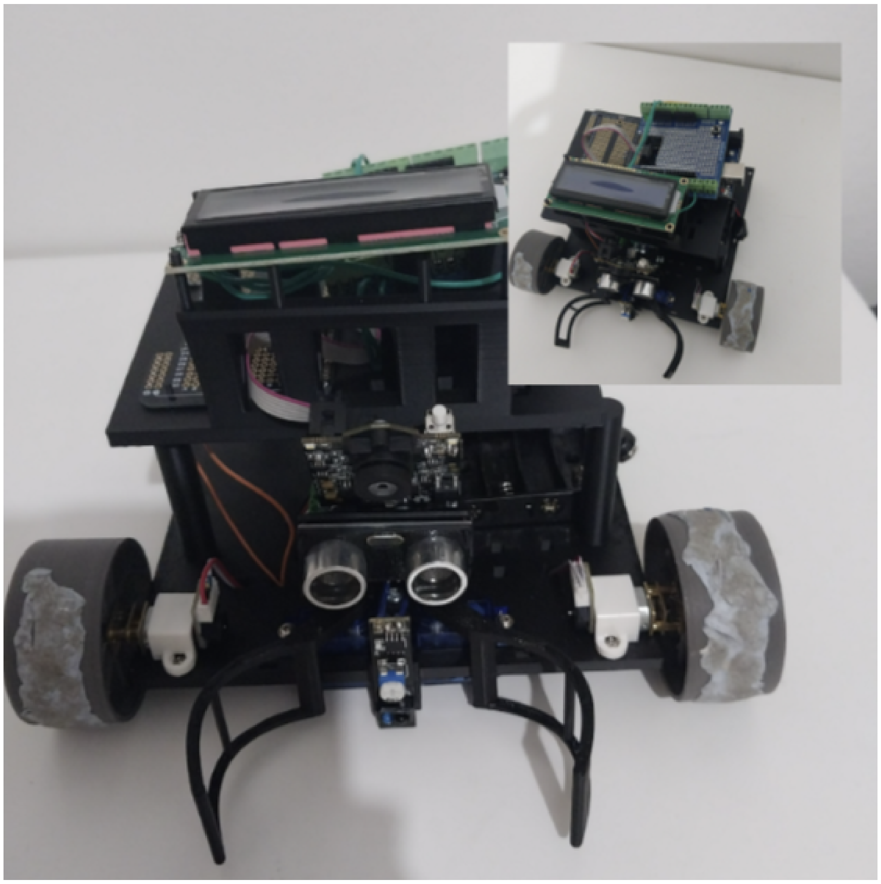


Figure 41: Assembled robot chassis without wiring. **Photo by Yusuf, built by me, Franklin, and Yusuf**

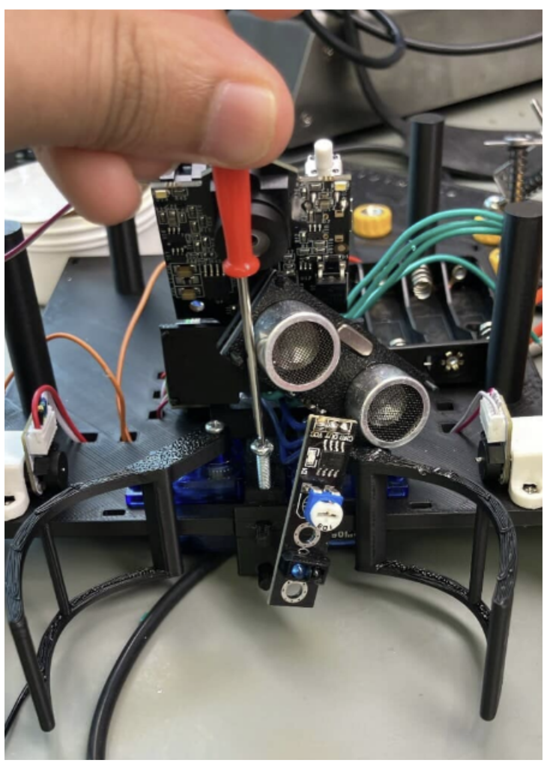


Figure 42: Servo motors mounted on bracket. **Photo by me**

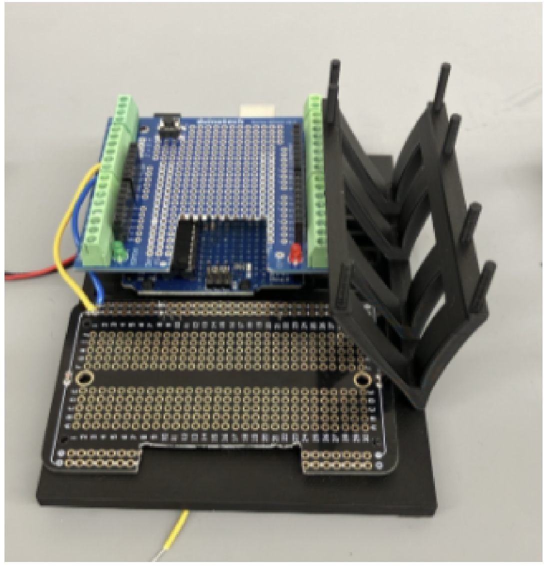


Figure 43: Top level of inner skeleton with LCD mount, Arduino shield, and perfboard. **Photo by me**

The mechanical assembly was done by the end of the session. One key coding decision I made for the final testing algorithm was the **bus isolation technique** for the ultrasonic sensor. The HC-SR04 uses digital GPIO pins for trigger and echo, but the problem is that when the I2C bus (used by the LCD) and the SPI bus (used by the Pixy2) are both active, the electrical noise on the shared lines can interfere with the ultrasonic timing measurements. Even a few microseconds of corruption in the echo pulse duration translates to several centimetres of error in the distance reading. To prevent this, **I wrote two functions** that shut down the I2C and SPI buses before taking an ultrasonic reading and then restore them afterwards. The LCD backlight and display are also turned off during the measurement to eliminate any I2C traffic. See **Figure** 44 for the bus isolation code.

```
// Bus isolation for ultrasonic precision
void isolateBusesForUltrasonic() {
  lcd.noBacklight();
  lcd.noDisplay();
  Wire.end();     // Shut down I2C
  SPI.end();      // Shut down SPI
  delay(100);
  pinMode(ECHO_PIN, INPUT);
  pinMode(TRIG_PIN, OUTPUT);
}

void restoreBusesAfterUltrasonic() {
  Wire.begin();
  lcd.init();
  lcd.backlight();
  pixy.init();
  delay(200);
}
```

Figure 44: Bus isolation technique: shuts down I2C and SPI during ultrasonic readings to prevent interference. **Coded by me**

1.5.2 Entry 14 (15/04)

On the 15th of April, Yusuf and I went to the Elec Makerspace from 2pm to 5pm to continue the electrical connections. Franklin assisted initially with remaining mechanical fitting before handing over to us. Wiring was not yet fully complete at the end of this session; some connections remained to be finished at the next build day before a full powered test could be run.

The first connections we made were the **clamp servos to the 4xAAA LiPo battery pack**. Servos run on a separate supply from the Arduino and motor driver to stop PWM switching current from causing voltage dips on the Arduino rail, which can trigger resets mid-run. Critically, we tied the **servo battery GND to the Arduino GND**. Without this shared reference, the PWM signal from the Arduino has no common voltage to switch against and the servos either jitter continuously or do not move at all [4]. We then **soldered the MKRVRS motor driver pinouts** to the Arduino, testing each with the multimeter continuity beep test before powering anything: DIR_A to pin 2, PWM_A to pin 3, DIR_B to pin 4, PWM_B to pin 5. See **Figure** 45 for the motor driver soldering.

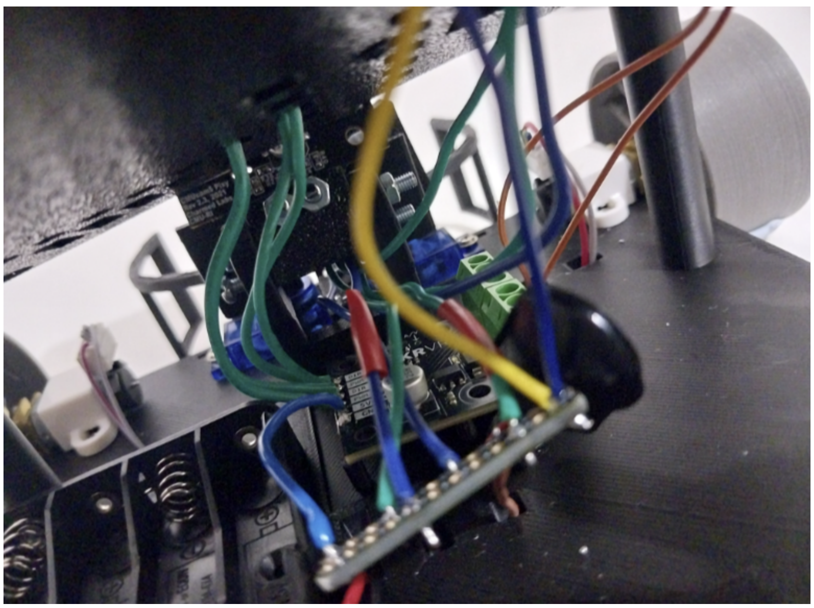


Figure 45: Motor driver pinouts soldered to Arduino with ultrasonic and IR common power/GND rail. **Photo by me**

Wiring the LCD to the perfboard revealed a fault: the VDD pin on the I2C backpack was not conducting. **We identified this with the multimeter** and desoldered that pin, switching to the secondary VDD pad. We also **created a dedicated power rail and GND bus on the perfboard** for the HC-SR04 and FC-51 IR sensor so both sensors share a single clean rail rather than individual wires back to the Arduino headers. See **Figures** 46 and 47 for the soldering progress.

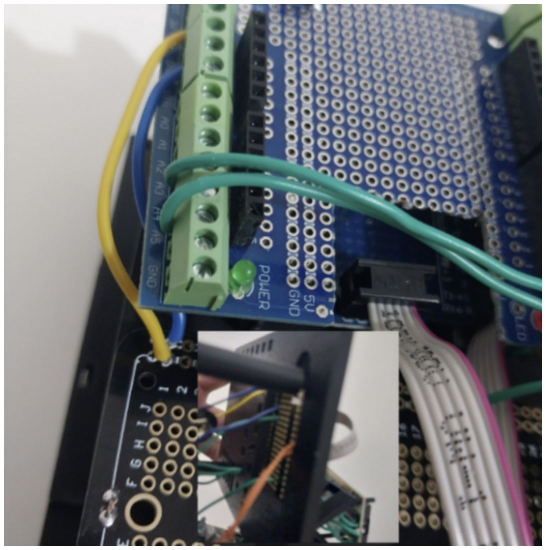


Figure 46: Perfboard and shield with wires soldered. **Photo by me**

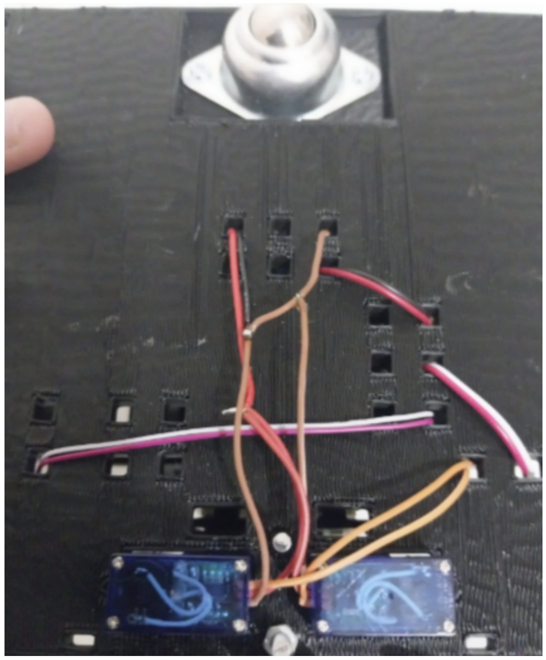


Figure 47: Wire routing through chassis holes; servo and motor driver wires soldered cleanly under chassis. **Photo by Yusuf**

1.5.3 Entry 15 (17/04)

On the 17th of April, the team met in the Elec Makerspace from 11am to 2pm. This was the final build session before the demonstration. Franklin, Bradman, and Yizhuo focused on the risk assessment and final report planning while Jesse trained the Pixy2 colour signatures for the new chassis configuration. Yusuf helped fix and tune the remaining electronics, and **I handled the remaining code refinements and individual component testing**.

The first priority was finishing the wiring connections left over from the 15th. Once those were done, **I tested each component individually** before combining them. The two sensors most at risk of interference due to their tight physical placement were the HC-SR04 ultrasonic and the FC-51 IR sensor, so these were tested first. The ultrasonic was confirmed working with the median filter and time jitter technique producing stable readings with no false triggers. The IR sensor was calibrated specifically to detect a ball sitting inside the clamp enclosure; since the sensor is enclosed, the detection threshold needed to be set for a very short range. Both sensors performed well in isolation.

The servo clamp was then tested for open and close angles. When the clamp was commanded to close, there was an initial jitter and a small jump before it settled into the closing motion. **I debugged this and identified the cause**: the solder joint connecting the servo to the battery negative terminal was not making proper contact, meaning the common GND connection was intermittent. This meant the servo's reference voltage was floating, which caused the brief erratic movement before the joint made momentary contact and the servo responded normally. Yusuf reflowed the solder joint, and after that the servo opened and closed cleanly with no jitter.

Once all individual components were confirmed working, **we connected everything together** as per the full system configuration. The complete wiring follows the block diagram from the Design Proposal with the addition of the LCD on the I2C bus [5]. See **Figure** 48 for the full system connectivity diagram.

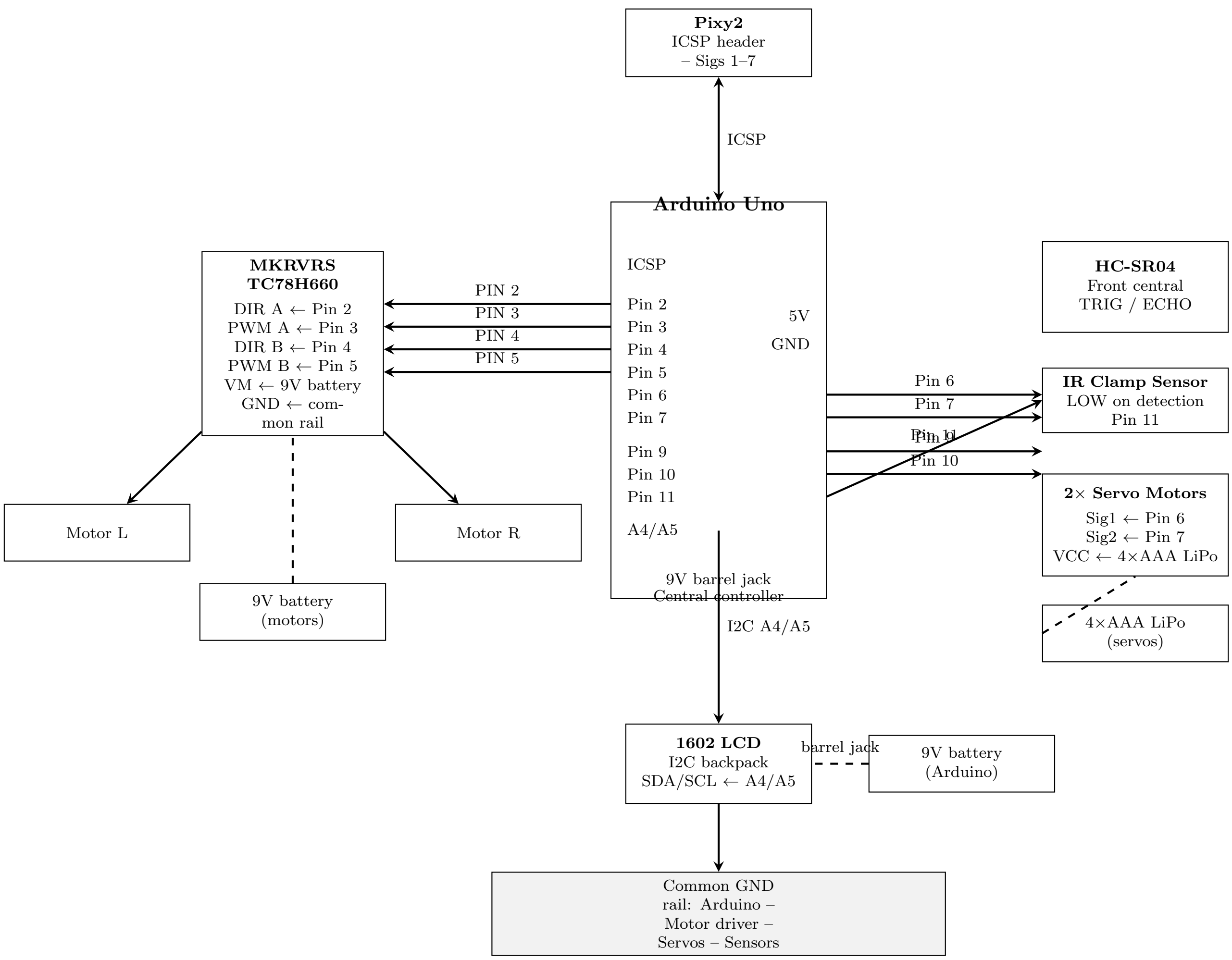


Figure 48: Full system connectivity diagram for final testing configuration, including LCD on I2C bus (A4/A5). **Made by me**

With the full system wired up, **I tested the sections of code that could be verified at this stage**. The key function to confirm was the fine positioning loop in `step4_finePosition()`, which uses the IR sensor's `ballInClamp()` reading to stop the robot as soon as the ball enters the clamp enclosure. This works by treating the IR output pin going LOW as a ball-present signal, then stopping all motor movement and exiting the loop. See **Figure** 49 for the `step4_finePosition()` code.

```
void step4_finePosition() {
  lcdShow("Step␣4:", "Fine␣positioning");

  while (!ballInClamp()) {
    int blockX = getBlockX(targetBallSig);

    if (blockX < Config::FRAME_CENTRE_MIN) {
      motorTurnLeft(Config::CRAWL_SPEED);
      delay(30);
      motorStop();
    } else if (blockX > Config::FRAME_CENTRE_MAX) {
      motorTurnRight(Config::CRAWL_SPEED);
      delay(30);
      motorStop();
    } else {
      motorForward(Config::CRAWL_SPEED);
      delay(50);
      motorStop();
    }
    delay(30);
  }
  motorStop();
}
```

Figure 49: `step4_finePosition()`: IR-guided fine positioning loop; crawls forward until ball triggers the clamp IR sensor. **Coded by me**

I also confirmed the two-phase base approach in `step9_approachBase()`, which first uses the Pixy2 block area to get within range, then isolates the I2C and SPI buses and switches to ultrasonic precision approach: slowing at 15cm and stopping at 5cm. See **Figure** 50 for the approach code.

```
void step9_approachBase() {
  lcdShow("Step␣9:", "Approach␣base");

  // Phase 1: Pixy area-guided approach
  while (getBlockArea(baseSig) < Config::BASE_AREA_THRESHOLD * 1.5) {
    motorForward(Config::APPROACH_SPEED);
    delay(50); motorStop(); delay(30);
  }

  // Isolate I2C/SPI for clean ultrasonic readings
  isolateBusesForUltrasonic();

  // Phase 2: Slow down at 15cm
  long dist = ultrasonicMedian();
  while (dist > Config::APPROACH_SLOW_CM || dist == 0) {
    motorForward(Config::APPROACH_SPEED);
    delay(50); motorStop(); delay(30);
    dist = ultrasonicMedian();
  }
  motorStop(); delay(300);

  // Phase 3: Crawl to 5cm
  while (dist > Config::APPROACH_STOP_CM || dist == 0) {
    motorForward(Config::CRAWL_SPEED);
    delay(30); motorStop(); delay(30);
    dist = ultrasonicMedian();
  }
  motorStop();
  restoreBusesAfterUltrasonic();
}
```

Figure 50: `step9_approachBase()`: two-phase precision base approach using Pixy2 area then ultrasonic with bus isolation. **Coded by me**

By the end of the session all individual components were verified, the servo jitter fault was found and fixed, and the fully integrated system was wired and running. Jesse had completed the Pixy2 signature training for all ball and base colours on the new chassis. The robot is ready for full end-to-end testing.

2 Teamwork

2.1 Week 5

2.1.1 Entry 1 (17/03)

On the 16th of March, our team held a meeting to discuss the plan heading into Flexi Week and compliance testing preparation. Jesse communicated the key priorities via Teams on the 17th, and I followed up with my own detailed roadmap announcement shortly after. The subgroups are communicating well with the overall leader, and Jesse is tracking progress on the team Gantt chart.

Time	16/03, 14:00 – 15:30
Location	Microsoft Teams (Online)
Attendance	All members were present.

Agenda	- Discuss compliance testing preparation and what needs to be done - Everyone must read the CT specification and rubric before Thursday's meeting - Collect availability for Flexi Week build days at UNSW - Franklin and Trishan to report on current robot progress and remaining work
Meeting Summary	- Jesse outlined the compliance testing timeline and key priorities for the team - We established that the robot needs to be ready for debugging by 26th March - Trishan posted a detailed roadmap on Teams breaking down the deliverables for both S1 and S2 - Franklin reported that chassis CAD work has begun on Fusion 360 - Yusuf began investigating power requirements and distribution for the electrical layout - The team agreed to evaluate S1 vs S2 feasibility at the subgroup leader meeting on 19th March
Decisions Made	- Robot must be hardware-built and basic code working by 26th March for debugging - Subsystem choice (S1 or S2) to be confirmed after subgroup leader meeting on 19/03 - All members must reply with Flexi Week availability before Thursday's meeting - Jesse tracking overall progress on the team Gantt chart

Action Items	• Everyone must read the CT specification and rubric before next meeting • Franklin and Trishan to provide a progress update on what has been completed and what remains • Trishan to evaluate S1 vs S2 feasibility and present findings at subgroup leader meeting • Franklin to continue chassis design on Fusion 360 • Yusuf to continue investigating power distribution requirements • All members to reply with Flexi Week availability

2.1.2 Entry 2 (19/03)

Jesse called a subgroup leader meeting on Teams to establish and assign the key tasks for compliance testing. The meeting was focused and productive; we came out with clear ownership of every deliverable.

Time	19/03, 17:00 – 18:00
Location	Microsoft Teams (Online)
Attendance	Jesse, Trishan, Franklin, Yusuf (subgroup leaders)
Agenda	• Establish and distribute tasks for compliance testing preparation • Discuss power requirements for motor driver and servos • Confirm subsystem choice direction based on Trishan's S1 vs S2 evaluation

Meeting Summary	• Jesse outlined the five key tasks and we assigned ownership as follows: • Task 1 (Power distribution and wiring layout): Yusuf and Yizhuo • Task 2 (Sensor integration and CT code): Trishan and Jesse • Task 3 (Chassis build and mechanical assembly): Franklin and Bradman • Task 4 (PixyCam and information display): Trishan • Task 5 (Motor driver setup and testing): Yusuf and Trishan • We identified that a separate 9V battery is needed for the MKRVRS motor driver due to the DC motor stall current exceeding the Arduino's power output • Power sources for servos and the Arduino itself are still TBD
Decisions Made	• Tasks distributed across subgroup leaders as listed above • Separate 9V battery confirmed for motor driver; servo and Arduino power TBD • Team leaning towards S1 for compliance testing based on current progress
Action Items	• Trishan to work on PixyCam integration, LCD display setup, and research Pixy2 SPI communication • Yusuf and Yizhuo to map out full power distribution and wiring layout • Franklin and Bradman to continue chassis build and begin mechanical assembly • Jesse and Trishan to begin writing CT demonstration code • Yusuf and Trishan to set up and test the MKRVRS motor driver with DC motors

2.1.3 Entry 3 (21/03)

Time	19/03 – 21/03
Location	Independent work / Microsoft Teams
Attendance	All members contributing independently
Agenda	• Each member progressing on their assigned compliance testing tasks from the subgroup leader meeting • Updates communicated via Teams
Meeting Summary	• Trishan completed the Pixy2 + LCD colour detection test for all three ball colours (red, green, blue) and communicated results on Teams with video evidence • Franklin designed and 3D printed prototype wheels, tested them and found the axle hole was too large; began a CAD redesign with Bradman. Also identified that wheel width was too large and adjusted accordingly • Yusuf figured out the MKRVRS motor driver pinout setup and set up an unsoldered configuration to test before committing to soldering • Jesse researched how to train PixyCam CCC colour signatures more effectively, including tweaking settings like overexposure for better detection reliability
Decisions Made	• Colour detection and LCD display confirmed as working for S1 compliance testing • Wheel redesign needed before next print; axle hole and width to be corrected • Motor driver pinout confirmed; soldering to follow once verified
Action Items	• Trishan to set up I2C backpack for the LCD once it arrives to free up Arduino pins • Franklin and Bradman to reprint corrected wheels • Yusuf to solder motor driver connections once testing is confirmed • Jesse to share PixyCam training findings with Trishan for integration

2.2 Week 6

2.2.1 Entry 4 (23/03)

The whole team came into the ElecEng Building and Makerspace for a full build session. We had a brief meeting at the start in the ElecEng Building Level 1 to align on priorities for the day, then split off to build, code, and assemble.

Time	23/03, 13:00 – 17:00
Location	ElecEng Building Level 1 / Makerspace
Attendance	All members were present.
Agenda	• Brief team alignment on the day's priorities • Begin physical assembly and integration of robot subsystems • Test key components: wheels, motor driver, PixyCam, and obstacle detection
Meeting Summary	• Franklin and Bradman brought in 3D printed wheels, laser cut a test plywood chassis, and 3D printed motor mounts which were screwed in for testing • Yusuf worked with Bradman and Franklin to set up MKRVRS motor driver pinouts and ran a successful motor test • Jesse trained PixyCam colour signatures with improved block filtering, autoexposure, and brightness settings • Trishan tested PWM tuning for motor movement (forward, backward, turning) and integrated IR sensor for obstacle stop/resume proof of concept
Decisions Made	• Test plywood chassis confirmed as functional for integration testing • Motor driver and DC motors verified as working together • IR obstacle detection with motor stop/resume logic confirmed as proof of concept

Action Items	• Jesse to continue researching overexposure settings and CCC training optimisation • Yusuf to solder the I2C backpack for the LCD • Yizhuo to write a logic structure for the base colour recognition navigation system • Trishan to continue writing and refining the R-G-B cycle code now that signatures are being trained well • Franklin and Bradman to continue designing the final PLA chassis on CAD

2.2.2 Entry 5 (25/03)

The team came into the ElecEng Building and Makerspace for another build session. Everyone was present and contributing to the physical assembly and integration of the robot.

Time	25/03, 13:00 – 17:00
Location	ElecEng Building / Makerspace
Attendance	All members were present.
Agenda	• Assemble the PLA chassis with all major components • Mount motors, Arduino, breadboard, sensor front plate • Solder I2C backpack and finalise circuit pinouts • Test PixyCam mounting angle and FOV
Meeting Summary	• Franklin and Bradman 3D printed the PLA chassis plate and mounted the DC motors, rear caster wheel, and motor driver on precision mounts • Top plate had Arduino UNO mounting points and breadboard installed • PixyCam and ultrasonic sensor front plate were mounted and the camera FOV and angle were tuned • Yusuf soldered the I2C backpack for the LCD and helped finalise the circuit pinout layout • Trishan, Jesse, Franklin, and Bradman all contributed to the physical assembly • After the session, Trishan took the robot home and rewrote the obstacle detection code using the ultrasonic sensor with a 20cm threshold; test was successful

Decisions Made	• PLA chassis confirmed as the build platform going forward (replacing plywood test chassis) • PixyCam mounting angle and FOV confirmed as suitable for ball and base detection • Ultrasonic sensor to be used for obstacle detection in compliance testing code
Action Items	• Trishan to integrate ultrasonic obstacle detection with PixyCam colour detection into a single codebase • Trishan to continue refining the R-G-B colour cycle navigation code • Franklin and Bradman to continue any remaining chassis refinements • Yusuf to verify I2C LCD is working with the soldered backpack • Team to aim for full debugging session by 26th March as planned

2.2.3 Entry 6 (26/03)

This was the big debugging day. The full team came in to finalise the wiring, test the obstacle detection on the floor, calibrate movement, and test the base colour recognition system. Jesse and I stayed late to sort out the base navigation.

Time	26/03, 13:00 – 21:00 (Trishan and Jesse stayed until 21:00)
Location	ElecEng Building / Makerspace / Arena
Attendance	All members present (13:00–17:00). Trishan and Jesse remained until 21:00.
Agenda	• Finalise wiring of all components to the Arduino and connect power supplies • Test obstacle detection on the floor with ultrasonic sensor • Calibrate rotation durations and PWM speeds for movement modes • Test base colour recognition system in the arena

Meeting Summary	• Yusuf soldered the I2C backpack onto the LCD and completed the full wiring layout of the robot • Yusuf will also solder the barrel jack wires to the 9V battery for a secure Arduino power connection • Bradman and Franklin mounted and assembled all components together securely on the PLA chassis • Full wiring completed: 9V LiPo for motors via MKRVRS, 9V barrel jack for Arduino • Obstacle detection tested on the floor at 100 PWM; robot stopped when foot was placed in front and resumed on removal • Trishan implemented the median filter and time jitter technique for ultrasonic crosstalk prevention • Movement calibrated: 80 PWM forward, 60 backward, 40 turning, 30 slow approach. 180°turn at 1900ms, 360°at 3800ms • Jesse and Trishan tested base colour recognition: raw RGB matching failed (2/20), CCC trained signatures worked every time • Trishan wrote the full S1 compliance testing algorithm
Decisions Made	• CCC-based base colour recognition confirmed as the navigation method (raw RGB abandoned) • PWM values and rotation durations locked in for compliance testing • Time jitter technique adopted for ultrasonic sensor to prevent crosstalk in multi-robot arena • S1 compliance testing algorithm finalised • Team agreed to come in on Sunday for a dry run and further debugging

Action Items	• Bradman and Franklin to evaluate whether a rear caster wheel or two linear rear wheels are better; decision to be made on Sunday • Trishan and Jesse to recalibrate movement and navigation in accordance with whichever rear wheel design is chosen • Trishan to refine and debug the full CT algorithm before Monday's test • Jesse to continue refining PixyCam CCC training for base signatures under different arena positions • Full team dry run scheduled for Sunday before Monday's compliance testing

2.2.4 Entry 7 (28/03)

The full team came in for a dry run on the Sunday before compliance testing. We treated this as a full rehearsal, testing every subsystem that would be demonstrated the next day.

Time	28/03, 12:00 – 18:00
Location	ElecEng Building / Makerspace / Arena
Attendance	All members were present.
Agenda	• Full dry run of S1 compliance testing demonstration • Test obstacle detection threshold and verify marking criteria compliance • Calibrate turns and verify movement consistency • Test base colour recognition and navigation to base

Meeting Summary	• Trishan calibrated the ultrasonic threshold distance for obstacle detection and ran tests against the CT marking criteria • Team identified that two rear wheels were causing inconsistent turns due to dragging during rotation • Franklin and Bradman removed the rear wheels and installed a single rear caster wheel for consistent on-the-spot rotation • Trishan recalibrated all PWM values and turn durations (5–10 PWM lower due to reduced friction) • Base colour recognition tested; issue found where PixyCam detected edge signature instead of centring on base • Trishan updated the base recognition logic to include a centring sweep before approach
Decisions Made	• Two rear wheels replaced with single caster wheel for consistent rotation • PWM values lowered by 5–10 to account for reduced friction • Base approach algorithm updated to centre signature in frame before moving forward
Action Items	• Full night's rest before compliance testing on Monday • Arrive early on Monday for final checks and arena familiarisation • Ensure all batteries are charged and backup batteries are available

2.3 Week 7

2.3.1 Entry 8 (30/03)

The whole team arrived at James Kirby Makerspace two hours before the CT session to run through final checks. Roles were clear from the start: electrical checked wiring, mechanical verified chassis and mounts, and the coding side was handled by Franklin and me. Jesse was present in an oversight and team leader capacity. The pre-CT session ran from 13:00 to 15:30, after which we moved into the actual compliance testing with the marker.

Time	30/03, 13:00 – 15:30 (pre-CT); CT session followed immediately after
Location	James Kirby Makerspace

Attendance	All members were present.
Agenda	• Lock in RGB colour detection cycle and verify LCD output for all three ball colours • Refine and test `centreOnBase()` sweep parameters in the arena • Run full movement and navigation sequence end-to-end • Proceed to CT session once all systems confirmed
Meeting Summary	• Yusuf checked all wiring and power connections; no faults found • Franklin and Bradman verified chassis and component mounting was secure • Trishan and Franklin ran the RGB colour detection cycle and tweaked signature parameters to ensure no false triggers • Trishan refined the `centreOnBase()` sweep-centre-approach function; tested repeatedly in arena until consistently on target • Full movement and navigation sequence run end-to-end and confirmed working • Team proceeded to the CT session with all systems locked and calibrated • Robot performed flawlessly during the CT demo; examiner David noted he was impressed with the performance • After CT, team read through the Design Proposal guide; Trishan assigned the bulk of Relevant Knowledge section plus some of Design Concept, and bibliography/-sourcing work with Yusuf

Decisions Made	• All algorithm parameters locked after pre-CT testing • `centreOnBase()` confirmed as the final navigation function for the CT demo • S1 compliance testing demo completed successfully • Design Proposal sections assigned: Jesse (Project Overview, Timeline), Trishan (Relevant Knowledge, part of Design Concept, Bibliography with Yusuf), Yusuf (Electrical, Bibliography), Franklin (Mechanical), Bradman (Testing Plan), Yizhuo (Documentation and Risk)
Action Items	• Trishan to document CT results and performance notes for Entry 9 • Team to debrief and discuss S2 preparation or post-CT improvements in next session • Team to read through the Design Proposal guide and sections assigned as follows: Jesse (Project Overview, Timeline), Trishan (Relevant Knowledge, part of Design Concept, Bibliography with Yusuf), Yusuf (Electrical, Bibliography), Franklin (Mechanical), Bradman (Testing Plan), Yizhuo (Documentation and Risk Assessment)

2.3.2 Entry 9 (02/04)

The full team met in the CompLabs to review Design Proposal progress before Sunday's submission. This was our chance to see what everyone had written and plan the final push.

Time	02/04, 12:00 – 15:00
Location	CompLabs
Attendance	All members were present.
Agenda	• Review each member's Design Proposal section progress • Identify gaps and areas needing refinement • Plan editing schedule for Sunday submission • Assign Saturday pre-check team for final edits

Meeting Summary	• Trishan presented Relevant Knowledge sections (PixyCam, ultrasonic, motor control), circuit diagram, and final testing algorithm flowchart • Jesse and Trishan set up shared Overleaf document for collaborative editing • Yusuf presented bibliography work; had done majority of sourcing • Franklin, Bradman, and Yizhuo presented their sections but drafts needed significant refinement • Team agreed that Jesse and Trishan would handle bulk of rewriting on Saturday • Saturday pre-check scheduled for Jesse, Trishan, and Yusuf to edit and refine document
Decisions Made	• Saturday pre-check team: Jesse, Trishan, Yusuf (editing and refinement) • Sunday submission deadline confirmed • Jesse and Trishan to rewrite and edit weaker sections so others can focus on final testing prep
Action Items	• Jesse and Trishan to edit and refine Design Proposal on Saturday • Yusuf to finalise bibliography formatting • Remaining team members to begin planning for final testing while Jesse and Trishan handle document • Final document review on Sunday before submission

2.3.3 Entry 10 (04/04)

Jesse, Yusuf, and I met virtually on Teams to finalise the Design Proposal. Most work was independent on Overleaf, with check-ins throughout the day.

Time	04/04, scattered sessions; main meeting 17:00 – 19:00
Location	Microsoft Teams (Virtual)
Attendance	Jesse, Trishan, Yusuf

Agenda	• Edit and format all Design Proposal sections • Create and embed figures, flowcharts, block diagrams • Finalise bibliography • Ensure Gantt chart reflects current progress
Meeting Summary	• Jesse and Trishan transferred Word drafts to LaTeX Overleaf • Trishan created TikZ flowcharts and block diagrams; formatted wall-following figure from Franklin's diagram • Yusuf and Trishan finalised bibliography in BibTeX • Trishan updated Gantt chart with current progress • Most sections needed only minor edits or paragraph rewrites • Document ready by Sunday morning; final spelling checks done
Decisions Made	• Design Proposal to be submitted Sunday after final checks • Figures to be created in TikZ for consistent formatting
Action Items	• Final spelling and grammar check Sunday morning • Submit Design Proposal by Sunday deadline

2.4 Week 8

2.4.1 Entry 11 (06/04)

The team met in the Main Library to review the final testing specification and plan the remaining work before the demonstration.

Time	06/04, 14:00 – 15:00
Location	Room 412, Main Library
Attendance	All members were present.
Agenda	• Progress check and read through final testing notification together • Establish action items for Thursday's meeting • Assign roles for final testing preparation

Meeting Summary	• Team read through final testing specification and made notes on marking criteria • Trishan made evaluation block chart for HD target • Trishan ordered and tested replacement LCD ($5) with HelloWorld print • Franklin made LCD mount placement diagram • PixyCam optimal position determined: 10cm height, 20–30 degree angle • Franklin and Bradman to design 3D printed mould with silicone curing for wheel covers • Roles assigned: Franklin (mechanical design), Yusuf (soldering), Bradman (assist), Jesse (oversight)
Decisions Made	• Team aims for HD in final testing • Wiring diagrams: Yusuf to complete • Robot paint (rusty colour): Jesse and Bradman • Risk Awareness & Mitigation: To be submitted separately by Thursday • All soldering to be done directly without header pins
Action Items	• Yusuf: wiring diagrams and soldering • Jesse and Bradman: robot paint (rusty colour) • Franklin: LCD mount, upper level (perfboard, Arduino), bottom level (motor driver, servo batteries) • Franklin and Bradman: 3D printed mould with silicone curing for wheel covers • Trishan: final testing algorithm code • Complete Risk Awareness & Mitigation by Thursday 9 April

2.4.2 Entry 12 (09/04)

The team met to confirm progress on action items and establish the Week 9 timeline.

Time	09/04, 14:00 – 15:00
Location	Room 412, Main Library

Attendance	All members were present.
Agenda	• Follow up on PixyCam positioning, wheel mould, LCD mount • Review final testing code • Establish Week 9 timeline and deadlines
Meeting Summary	• PixyCam mounting angle confirmed: 10cm height, 20–30 degree angle • Franklin and Bradman: 3D printed wheel mould for silicone, LCD mount printed • Trishan completed final_testing.ino with full ball pickup and sorting algorithm • Week 9 timeline established: Mon (code/CAD/diagrams), Tue-Wed (assemble), Thu (train Pixy2), Fri-Sun (test/paint) • Electrical subgroup reminded to communicate with Franklin before circuit diagram • Batteries and spare components to be confirmed and purchased
Decisions Made	• All subsystem code to be completed by Tuesday-Wednesday (hard deadline) • Circuit diagram needed urgently for Franklin's CAD work • Paint finish (rusty colour for Wall-E design) to be done after assembly
Action Items	• Trishan: attend Monday Lab/Freebuild for components needed for coding • Franklin: complete chassis CAD, coordinate with Yusuf on circuit diagram placement • Yusuf: complete circuit diagram, check datasheets, communicate with Franklin • Jesse and Bradman: decide on wheel covers and paint for Wall-E design • All: confirm and purchase batteries + spare components

2.5 Week 9

2.5.1 Entry 13 (13/04)

The team met to brief everyone before assembly. Subgroup leaders handled build and code; others prepped final report.

Time	13/04, time varies (later than planned)
Location	Room 412, Main Library
Attendance	All members were present.
Agenda	• Brief team on assembly and code • Present algorithm and final_testing.ino draft • Assemble chassis once prints ready
Meeting Summary	• Trishan presented algorithm and walked through final_testing.ino • Yusuf explained soldering plan for perfboard • Franklin showed CAD models: clamps, base plate, top plate, LCD mount, Wall-E casing • Subgroup leaders brainstormed pinouts, wire routing, servo bracket placement • Franklin and Yusuf assembled chassis with components on precision mounts
Decisions Made	• Wednesday: Yusuf and Trishan complete electrical connections • Friday/Saturday: Hand off to Trishan for testing and debugging
Action Items	• Wednesday: Yusuf and Trishan solder and wire all components • Friday/Saturday: Trishan receives robot for testing

2.5.2 Entry 14 (15/04)

Trishan and Yusuf soldered and wired all electrical components. Franklin assisted initially with mechanical assembly.

Time	15/04, 14:00 – 17:00
Location	Elec Makerspace
Attendance	Trishan, Yusuf, Franklin

Agenda	• Solder all electrical components • Wire motor driver, servos, LCD, sensors to Arduino • Test connections with multimeter
Meeting Summary	• Soldered clamp servos to 4AAA LiPo batteries with common GND to Arduino to filter PWM spikes • Soldered motor driver pinouts to Arduino and motors; tested with multimeter beep test • Wired LCD to Arduino and perfboard • Desoldered LCD VDD pin and used secondary VDD after identifying fault with multimeter • Created power rail and ground section on perfboard for ultrasonic and IR sensor
Decisions Made	• Common GND between servo batteries and Arduino to prevent PWM noise • Perfboard power rail for sensors for easier setup
Action Items	• Continue wiring if needed • Hand off to Trishan for testing on Friday/Saturday

2.5.3 Entry 15 (17/04)

The final build session before demonstration day. Roles were split clearly: Jesse trained Pixy2 signatures, Yusuf tuned electronics, Trishan coded and tested components, and Franklin, Bradman, and Yizhuo worked on the risk assessment and final report planning.

Time	17/04, 11:00 – 14:00
Location	Elec Makerspace
Attendance	All members were present.
Agenda	• Finish remaining wiring connections from previous session • Test ultrasonic, IR, and servo individually before full integration • Debug any wiring or code issues found during testing • Connect all components together for full system test • Jesse to train Pixy2 signatures on new chassis configuration

Meeting Summary	• Remaining wiring connections completed by Yusuf and Trishan • Ultrasonic tested: median filter and time jitter confirmed stable, no false triggers • IR sensor calibrated for ball detection inside the clamp enclosure • Servo clamp tested: initial jitter and jump identified; traced to poor solder joint on battery negative terminal • Yusuf reflowed the solder joint; servo open/close confirmed working cleanly after fix • All components connected together as per full system configuration • Trishan tested `step4_finePosition()` and `step9_approachBase()` code sections • Jesse completed Pixy2 CCC signature training for all ball and base colours • Franklin, Bradman, Yizhuo progressed risk assessment and final report planning
Decisions Made	• All individual components confirmed working; proceed to full end-to-end testing • Pixy2 signatures locked for final testing configuration • Robot handed to Trishan for full algorithm testing runs
Action Items	• Trishan to run full end-to-end algorithm tests and debug as needed • Franklin and Bradman to finalise any remaining mechanical refinements • Yizhuo to complete risk assessment documentation • Team to regroup before demonstration day for final checks

3 Reflections

3.1 Week 5

3.1.1 Entry 1 (17/03)

What am I doing well?

- I think starting the S1 vs S2 evaluation early and getting it out to the team quickly was a good move. Instead of waiting until the subgroup leader meeting to start thinking about it, I assessed our progress against both options and put together a flowchart to structure the comparison. This gave the team something concrete to discuss rather than going in blind.

What is our team doing well?

- Communication between the subgroups and Jesse as overall leader is pretty strong right now. Jesse's Teams announcement set clear expectations for everyone heading into Flexi Week, and I was able to follow up with a technical roadmap shortly after which shows the leadership structure is working. Jesse is also actively tracking progress on the team Gantt chart which keeps everyone accountable.

Why were the task(s) important?

- Evaluating S1 vs S2 early matters because the subsystem choice determines what hardware we need to build, what code needs writing, and what constraints we have to design around. If we made this call too late we'd either be scrambling to prepare the wrong stuff or switching direction mid-build. By laying out the requirements and our current progress in a flowchart, I gave the subgroup leaders what they need to make a proper decision at the 19th meeting.
- Jesse's announcement about reading the CT rubric and submitting Flexi Week availability also makes sure no one shows up unprepared and we can make the most of our build time at uni during the break.

What could I have done better?

- Looking back, I think a LucidSpark mindmap would have been better than a flowchart for this evaluation. A mindmap would let me branch out the sub-deliverables for each option and make it easier for other team members to add their own thoughts. The flowchart worked well for my own thinking but it is not as collaborative.

Is the progress in alignment with schedule?

- Yes, we are in a good spot heading into compliance testing prep. The EDP is done, key decisions have been discussed, and the roadmap for the next few weeks is laid out. The fact that we are already evaluating subsystem options and assigning early build tasks during Flexi Week means we are not wasting time between the EDP and compliance testing.

What have I learned and what do I need to learn?

- I learned that going from a presentation phase to a build phase needs a different kind of planning. The EDP was all about research, concepts, and communication; compliance testing is about physical deliverables and actually demonstrating functionality. Reading through the CT rubric properly helped me see exactly what marks are on the table and what constraints we have to work within (no laptop for S1, no hardcoding for both).

- I need to learn how to hook the PixyCam's colour detection output into the motor driver so the robot can respond to detected colours on its own. This is going to be essential no matter which subsystem we go with. I also need to figure out how to do LED-based output for S1 since serial monitor is not allowed.

- I communicated the detailed roadmap to the whole team via Teams, breaking down what each subsystem option needs and what we should prioritise. This set things up for the subgroup leader meeting on the 19th.

3.1.2 Entry 2 (19/03)

What am I doing well?

- I took the initiative straight after the subgroup leader meeting to start working on my assigned tasks rather than waiting. Discovering that the Pixy2 cannot run line tracking and CCC mode at the same time was a useful finding that I got from reading through the documentation on my own, and it directly shaped our navigation approach for compliance testing. On top of that, I got the LCD wired up and displaying output which is a solid step towards meeting the information display requirement for S1.

What is our team doing well?

- The subgroup leader meeting was efficient and well run by Jesse. We came out of it with clear task ownership and everyone knew exactly what they needed to work on. The fact that we identified the separate 9V battery requirement for the motor driver early means we will not run into power issues later during integration. The leadership structure is continuing to work well with subgroup leaders taking responsibility for their areas.

Why were the task(s) important?

- Figuring out the Pixy2's limitation with line tracking and CCC mode was critical because it rules out one of the two navigation methods for returning to base. If we had not discovered this now, we could have wasted time trying to implement line following only to find out it conflicts with our colour detection code. Now we know base colour recognition is the way to go.

- Getting the LCD working early is important because S1 does not allow serial monitor, so without an alternative display method we would lose marks on the information display criterion. Even though the direct wiring uses too many pins, at least I now know the LCD itself works and the I2C backpack will fix the pin issue.

What could I have done better?

- I should have checked whether I had an I2C backpack before wiring the LCD directly. That would have saved me time on a connection method I knew was not going to be practical for the final build due to pin limitations. I also could have tested the LEDs more thoroughly before giving up on them; they might have been workable with a shift register or multiplexer but I did not explore that.

Is the progress in alignment with schedule?

- Yes, things are moving well. Tasks are distributed, the PixyCam limitation has been identified early, the LCD is partially working, and power requirements are being mapped out. We are on track to have the key components ready for integration during Flexi Week.

What have I learned and what do I need to learn?

- I learned that the Pixy2 uses SPI protocol via the ICSP header for two-way communication with the Arduino, and that it cannot run line tracking and CCC mode at the same time. I also learned how to wire a 1602 LCD directly to the Arduino using a potentiometer for contrast, though this method is not practical for our build due to pin usage.

- I need to learn how to set up the I2C backpack for the LCD once it arrives and how to integrate the Pixy2's block data into the main control loop so we can use colour detection to drive motor commands. I also need to work with Yusuf to figure out the servo power source.

- I communicated the Pixy2 line tracking limitation to the subgroup leaders, which confirmed that base colour recognition is our navigation strategy. I also shared the LCD progress so the team knows the display side of things is being handled.

3.1.3 Entry 3 (21/03)

What am I doing well?

- Getting the Pixy2 and LCD working together end-to-end is probably the most significant individual contribution so far in the build phase. It proves that our colour detection pipeline actually works; the camera detects a ball, the Arduino reads the signature, and the LCD displays the correct action. This is not just a component test anymore, it is a working subsystem chain that directly maps to marks in the CT rubric.

What is our team doing well?

- Everyone is making progress on their tasks independently without needing to be chased. Franklin finding and fixing the wheel issues early through physical prototyping is exactly the kind of iterative approach we need. Yusuf getting the motor driver pinout sorted before soldering shows good caution, and Jesse's research into PixyCam settings will help refine what I have already set up. The team is working in parallel and communicating updates on Teams which keeps momentum going.

Why were the task(s) important?

- The Pixy2 + LCD test is directly tied to three CT marks: colour detection system exists (1 mark), robot identifies specified colour (1 mark), and robot produces a display response for the identified colour (1 mark). Having this working now gives us confidence that a significant chunk of S1 is already sorted and we can focus on the remaining criteria like obstacle detection and navigation.

- Franklin's wheel prototyping is also important because without functional wheels the robot literally cannot move, and catching the axle and width issues now rather than during integration saves us a lot of time.

What could I have done better?

- I should have waited for the I2C backpack to arrive before doing the full LCD test. I ended up using 6 pins on the Arduino which I know we cannot afford in the final build, so I will have to redo the wiring once the backpack comes in. It was not wasted effort since I confirmed the LCD works, but it would have been more efficient to do it once with the right setup.

Is the progress in alignment with schedule?

- Yes, and I would say we are slightly ahead. The colour detection and display system is working, wheels are being iterated on, motor driver pinout is confirmed, and PixyCam settings are being refined. All of this is happening during Flexi Week as planned, and we are on track for the 26th March debugging target.

What have I learned and what do I need to learn?

- I learned how to use the `LiquidCrystal` library in the Arduino IDE to print text on the LCD, including cursor positioning with `lcd.setCursor()` and clearing the display between detections. I also learned first hand how quickly Arduino pins get used up when you wire components directly, which reinforced why the I2C backpack is necessary.
- I need to learn how to set up and code with the I2C backpack (`LiquidCrystal_I2C` library) once it arrives. I also need to start thinking about how to structure the main control loop so that colour detection, obstacle detection, and motor commands all run together without conflicts.
- I communicated the full test results to the team on Teams with video evidence, so everyone can see that the colour detection and display pipeline is confirmed working. This gives the team confidence heading into integration.

3.2 Week 6

3.2.1 Entry 4 (23/03)

What am I doing well?

- I think the biggest thing I did well today was taking the test chassis after the team had set it up and actually getting the motors and IR sensor working together in code. That is the first time we have had a robot that can move and react to its environment, even if it is just a basic stop/resume on obstacle detection. Tuning the PWM values for different directions also gave me a feel for how the motors behave which will be useful when we write the full navigation code.

What is our team doing well?

- The whole team showed up and everyone was productive for the full four hours. There was no standing around or waiting on each other; Franklin and Bradman handled the mechanical side, Yusuf got the motor driver going, Jesse refined the PixyCam, and then I picked up from there with the coding and integration. This is the kind of parallel workflow that gets things done quickly and it felt like a proper build session for the first time.

Why were the task(s) important?

- Having the motors respond to sensor input through the Arduino is a fundamental requirement for compliance testing. Without this, the robot is either moving blindly or not moving at all. The IR obstacle test proves the control loop works: sensor reads data, Arduino processes it, motor driver responds. This same pattern will scale up to the ultrasonic sensor and PixyCam integration.
- The test chassis itself is also important because it gives us a physical platform to test code on rather than just simulating or testing components in isolation on a breadboard.

What could I have done better?

- I should have tested the ultrasonic sensor for obstacle detection as well rather than just the IR sensor. The ultrasonic gives us distance data which is more useful for the actual CT demo where the robot needs to stop and resume at specific thresholds. The IR was a quicker proof of concept but I could have pushed further while I had the setup ready.

Is the progress in alignment with schedule?

- Yes, we are right on track. The test chassis is assembled, motors are confirmed working with the driver, colour signatures are being trained, and I have a basic obstacle detection loop running. The 26th March debugging target is looking achievable at this rate.

What have I learned and what do I need to learn?

- I learned how to tune PWM values on the MKRVRS motor driver to control speed and direction for differential steering. I also learned that the IR sensor's binary output (HIGH/LOW) makes it very simple to integrate with motor stop/resume logic, which is why it was a good starting point before moving to the ultrasonic.
- I need to learn how to integrate the ultrasonic sensor's distance readings into the motor control loop with proper thresholds, and I need to start working on the R-G-B colour cycle code so the robot can target specific colours in sequence. I also need to figure out how the PixyCam's X-coordinate data can be used to steer the robot towards a detected ball.
- I communicated the obstacle detection test results and PWM tuning findings to the team, and shared the test video so everyone could see the motors responding to the IR sensor in real time.

3.2.2 Entry 5 (25/03)

What am I doing well?

- I think the most valuable thing I did today was taking the assembled robot home and upgrading the obstacle detection from the IR sensor to the ultrasonic sensor with an actual distance threshold. The IR version was binary (object or no object) but the ultrasonic version gives us real distance data at 20cm, which is what we actually need for compliance testing. Taking initiative to do this outside of the group session meant we did not lose any time.

What is our team doing well?

- Today felt like everything clicked. The mechanical subgroup had the PLA chassis printed and assembled with motor mounts, the electrical side had the motor driver and I2C backpack sorted, and then I could pick up from there and start integrating code on a real chassis rather than a breadboard setup. Everyone contributed to the assembly and nobody was idle. Yusuf soldering the I2C backpack is going to save us a lot of pin headaches going forward.

Why were the task(s) important?

- Having a fully assembled chassis with all components mounted is a major milestone. Up until now we have been testing things in isolation or on the plywood test chassis. Now we have the actual robot platform that we will be demonstrating at compliance testing. The ultrasonic obstacle detection code is also directly tied to CT marks for obstacle detection (2 marks in S1).

- The PixyCam FOV and angle check was also important because if the camera is mounted at the wrong angle, it either misses balls on the ground or cannot see the base colour plates. Getting this right now means we do not have to debug camera issues during compliance testing.

What could I have done better?

- I could have started integrating the ultrasonic and PixyCam code into a single sketch rather than keeping them separate. Right now I have the obstacle detection working and the colour detection working but they are in different codebases. Merging them sooner would give us more time to debug the combined system.

Is the progress in alignment with schedule?

- Yes, we are right on target. The chassis is assembled, sensors are mounted, obstacle detection is working with the ultrasonic, and the PixyCam FOV is confirmed. We are hitting the 26th March debugging target that we set back in Week 5 which is a good sign that the planning from the subgroup leader meeting paid off.

What have I learned and what do I need to learn?

- I learned how to implement the ultrasonic sensor with a distance threshold in the motor control loop, which is more practical than the IR's binary detection. I also saw first hand how important the PixyCam mounting angle is for getting reliable detections; even a few degrees off changes what the camera can see significantly.

- I need to learn how to merge the obstacle detection and colour detection code into one unified sketch that runs both systems in the main loop without them interfering with each other. I also need to start coding the base colour recognition logic for the "return to base" CT requirement.

- I communicated the ultrasonic test results to the team via Teams and shared the video evidence. The team now knows that obstacle detection is working on the actual robot and we are ready to start combining subsystems.

3.2.3 Entry 6 (26/03)

What am I doing well?

- This was by far my most productive day on the project. I implemented the median filter, came up with the time jitter technique on my own to handle multi-robot ultrasonic interference, calibrated all the PWM and rotation values, and wrote the full S1 algorithm. Staying late with Jesse to debug the base colour recognition was also important; we could have left it for another day but sorting it now means we are not scrambling before Monday's test.

What is our team doing well?

- The team hit the 26th March debugging target that we set back in Week 5, which shows the planning paid off. Everyone contributed to the wiring and assembly, and the fact that Jesse stayed late with me to work on the base navigation shows real commitment. The whole build from chassis assembly to a working obstacle-detecting, base-recognising robot happened in essentially three build sessions across the week, which is solid progress.

Why were the task(s) important?

- The time jitter technique is important because without it, our ultrasonic readings could be completely unreliable in the multi-robot arena. A false short reading would cause the robot to stop randomly, which would fail the smooth operation criterion. The median filter alone is not enough since it handles random noise but not systematic crosstalk from other robots' sensors.
- Calibrating the rotation and PWM values is essential because the compliance testing algorithm depends on accurate 180°and 360°turns. If these are off, the robot will miss the base during its scan or overshoot during approach. And figuring out that CCC training works for base recognition while raw RGB does not saved us from building our algorithm on an unreliable foundation.

What could I have done better?

- Jesse and I should have tested the base colour recognition earlier in the week rather than leaving it to the evening of the 26th. If it had not worked at all, we would have been in trouble with only a few days before compliance testing. We got lucky that the CCC approach worked, but the risk was unnecessary.

Is the progress in alignment with schedule?

- Yes, we hit the debugging target right on the day we planned. The robot has obstacle detection, calibrated movement, base colour recognition, and a full S1 algorithm. There is still debugging and refinement to do before Monday, but the core functionality is there which is exactly where we wanted to be.

What have I learned and what do I need to learn?

- I learned that raw RGB signature matching is unreliable under changing lighting conditions and that CCC trained signatures are far more robust for base identification. I also learned the value of the time jitter technique for preventing ultrasonic crosstalk; this was something I came up with by thinking about the arena conditions rather than just testing in isolation.

- I need to do a full end-to-end test of the S1 algorithm in the arena to make sure all the steps flow together without issues. I also need to make sure the colour detection and display (LCD) is properly integrated into the algorithm so we pick up the information display marks during CT.

- I communicated all findings to the team: the calibrated values, the base recognition approach, and the full algorithm. Jesse and I stayed late and pushed through the base navigation debugging together, which was a strong collaborative effort.

3.2.4 Entry 7 (28/03)

What am I doing well?

- I think the most important thing I did today was identifying the rear wheel issue and solving it through the caster wheel change. I could have just accepted that turns were inconsistent and tried to compensate in code, but I realised that the mechanical design was the root cause. Getting Franklin and Bradman to install the caster and then recalibrating everything meant we ended the day with a robot that can turn reliably, which is essential for the 360° scan in the S1 algorithm.

- Updating the base recognition logic to include the centring step was also critical. Without it, the robot would approach the base at an angle and potentially miss it entirely during the CT demo. Catching this during the dry run meant we could fix it before the real thing.

What is our team doing well?

- The team is good at pivoting quickly when something is not working. Once we identified the rear wheel issue, Franklin and Bradman immediately located the centrepoint and bolted on the caster wheel without needing to be told exactly what to do. Everyone stayed focused on the dry run objective and we did not waste time on things that were already working; we just tested, identified problems, fixed them, and moved on.

Why were the task(s) important?

- The dry run was our last chance to catch issues before the real compliance test. The rear wheel problem would have caused the robot to fail the navigation criteria because inconsistent turns mean the robot cannot reliably face the base or scan correctly. The base centring issue would have caused the robot to miss the base entirely or approach at a bad angle. Finding and fixing both on Sunday meant we went into Monday's CT with a working robot rather than hoping things would work on the day.

- Calibrating the ultrasonic threshold was also important because the obstacle detection marks depend on the robot stopping at the right distance. Too far and it might not detect smaller obstacles; too close and it risks collision.

What could I have done better?

- I should have tested the base recognition earlier in the week rather than leaving it to the day before CT. If the centring issue had been harder to fix, we could have run out of time. We got lucky that the fix was straightforward, but the risk was unnecessary given how close we were to the deadline.

Is the progress in alignment with schedule?

- Yes, the dry run happened exactly when we planned it, on the Sunday before Monday's compliance test. We found and fixed the key issues, and the robot is now mechanically sound and has updated logic for base navigation. We are as ready as we can be.

What have I learned and what do I need to learn?

- I learned that mechanical issues can have a bigger impact on code than I expected. The rear wheels dragging was causing turn inconsistencies that no amount of code calibration could fully fix. Changing the hardware made the software calibration work properly. I also learned that PixyCam signature detection is position-sensitive; just because a signature is detected does not mean the robot is facing it correctly.

- I need to remember to bring backup batteries for compliance testing and make sure everything is charged the night before. The last thing we need is a low battery affecting PWM performance during the demo.

- I communicated the centring logic fix and the new calibrated values to the team so everyone knows the final configuration going into compliance testing.

3.3 Week 7

3.3.1 Entry 8 (30/03)

What am I doing well?

- Coming in two hours early was the right call. It gave Franklin and me enough time to properly test and refine the `centreOnBase()` function under actual arena conditions rather than just trusting the version from the dry run. The pre-CT session caught a few parameter issues that would have caused problems during the real demo, so having that buffer mattered. Getting the RGB cycle locked before we went in was also important; that part of the demo needed to be clean and reliable.

What is our team doing well?

- The subgroup structure worked really well on the day. Electrical and mechanical handled their checks in parallel while Franklin and I focused entirely on the code and algorithm. Jesse managed the overall flow and kept things on track. Nobody got in each other's way and we were ready well before the CT session started.

Why were the task(s) important?

- The pre-CT session was the last chance to catch anything that could cause a failure during the actual demo. Getting the colour detection cycle locked and the base sweep working reliably in the real arena was critical because the examiner would be watching every step. If the `centreOnBase()` function had drifted or the RGB cycle had false triggered, we would have lost marks on criteria we had otherwise solved. The robot needed to perform consistently, not just once.

What could I have done better?

- I could have written the `centreOnBase()` function with a wider centre tolerance initially so it required less fine-tuning on the day. The 120 to 196 range worked but it took several test runs to confirm the parameters were tight enough without being too sensitive. Building in a configurable tolerance from the start would have made the pre-CT session faster.

Is the progress in alignment with schedule?

- Yes. Every target from the Week 5 roadmap was hit: chassis and wiring by the 26th, full algorithm written, dry run on the 28th, and CT completed on the 30th. The robot performed flawlessly on the day according to the examiner, which is the best possible outcome at this stage.

What have I learned and what do I need to learn?

- I learned that testing in the actual environment matters more than I initially gave it credit for. The arena lighting and floor surface do affect sensor behaviour, and what worked at home needed tweaking on-site. I also learned that splitting pre-CT responsibilities clearly by subgroup made the whole session much more efficient.
- Looking ahead, I need to start thinking about whether we want to attempt S2 and what that would involve for the code side, particularly the servo clamp and ball pickup logic.
- I also need to start on my Design Proposal sections: the bulk of Relevant Knowledge, part of Design Concept, and the bibliography with Yusuf. Jesse is handling Project Overview and Timeline, Yusuf is doing Electrical, Franklin is doing Mechanical, Bradman is doing the Testing Plan, and Yizhuo is covering Documentation and Risk Assessment.
- I communicated the final locked parameters and the CT outcome to the team after the session. Everyone was pleased with how it went.

3.3.2 Entry 9 (02/04)

What am I doing well?

- I got my sections done early enough to present properly. The TikZ circuit diagram took time but was worth it, and the final testing algorithm flowchart gave us something concrete for the next phase. Setting up Overleaf with Jesse meant we could edit efficiently on Saturday.

What is our team doing well?

- The progress check was honest; we recognised most sections needed refinement. Having Jesse and me handle the rewriting let others shift to final testing prep while we focused on making the document submission-ready.

Why were the task(s) important?

- The Design Proposal is worth significant marks. Drafting and reviewing before the weekend gave us buffer time for edits. The circuit diagram and algorithm flowchart also document what we actually built and coded.

What could I have done better?

- I could have started bibliography work earlier instead of leaving most to Yusuf. He did solid work, but I was assigned that section too.

Is the progress in alignment with schedule?

- Yes, every section has a draft and we have a plan to refine by Sunday. Saturday editing gives us a realistic shot at a clean submission.

What have I learned and what do I need to learn?

- I learned that TikZ diagrams in Overleaf are time-consuming but pay off in clarity. I also learned that dividing work by having two people focus on refinement while others prepare for testing is efficient.
- I need to learn how to write technical documentation that is concise but complete; some Relevant Knowledge sections were too verbose.
- I communicated the Overleaf link and editing plan to the team.

3.3.3 Entry 10 (04/04)

What am I doing well?

- I handled the LaTeX formatting and TikZ figures efficiently. Transferring Word drafts to Overleaf could have been messy but keeping figures consistent in TikZ made the final document look professional. The Gantt chart update ensured our timeline was accurate.

What is our team doing well?

- The virtual editing worked well because we divided tasks clearly. Jesse handled sections he knew best, Yusuf focused on bibliography, and I did figures and formatting. We did not step on each other's work.

Why were the task(s) important?

- The Design Proposal is assessed on clarity and presentation as well as content. Figures that are consistent and properly formatted make the document easier to read and show professionalism. The bibliography also needs to be complete and correctly cited.

What could I have done better?

- I could have asked Franklin for the wall-following diagram earlier instead of receiving it during the editing session. That would have saved some time.

Is the progress in alignment with schedule?

- Yes, we hit the Sunday submission deadline. The document was fully formatted and checked by Sunday morning.

What have I learned and what do I need to learn?

- I learned how to manage a collaborative Overleaf document with multiple editors. TikZ figure creation is also something I got faster at through practice.
- I need to continue refining technical writing to keep it concise; some sections were still too long after editing.
- I communicated the final figures and formatting to Jesse so he could verify everything was in place before submission.

3.4 Week 8

3.4.1 Entry 11 (06/04)

What am I doing well?

- Ordering and testing the replacement LCD the same day we read the final testing spec was the right move. It cost $5 and took less than an hour to confirm it worked with a HelloWorld print, but it meant Franklin could design the mount around actual hardware rather than working off assumptions. I also think the HD evaluation block chart was useful; it forced me to break down what the robot actually needs to do to score well rather than just broadly thinking "we need to sort balls." Having that visual made the meeting more focused.

What is our team doing well?

- Reading the final testing spec together as a group rather than individually was a good call. Everyone picked up on different things; for example, Yusuf flagged the circuit diagram requirement and Franklin immediately noted implications for CAD layout that I would not have caught. The role assignments also came out of that discussion naturally, which made it feel less like top-down delegation and more like everyone knowing what they needed to own. That kind of shared understanding tends to result in fewer gaps later.

Why were the task(s) important?

- The final testing specification is the single most important document for this phase. If any subgroup misunderstands a criterion, we could spend a week building or coding something that does not actually get marked. Going through it together and writing notes meant we had a shared reference to work from. On the technical side, confirming the PixyCam height at 10cm and 20–30 degrees is important because the mechanical design has to accommodate it; if we locked the camera mount at the wrong angle, it would be very hard to adjust once everything is assembled.

What could I have done better?

- I should have ordered the LCD earlier, ideally before the Design Proposal submission. We knew we needed it for the final build and waiting until Week 8 meant Franklin had to design the mount in the abstract without having the physical component to check dimensions against. It worked out, but it added an unnecessary dependency. I also could have drafted the HD evaluation chart before the meeting and sent it to the team in advance so we could walk in with a clearer starting point rather than making it during the session.

Is the progress in alignment with schedule?

- Broadly yes, but we are closer to the deadline than I would like. We have clear action items and role assignments, which is good, but the final build is more complex than the CT demo and I am conscious that Week 9 is going to be very dense. The Week 9 timeline we are planning to set on Thursday will be the real test of whether we have enough buffer.

What have I learned and what do I need to learn?

- I learned that going through an assessment rubric as a group surfaces things that get missed when people read it alone. It is a straightforward thing but it genuinely changed how we structured the action items coming out of this meeting. I also confirmed that the PixyCam's detection reliability is very sensitive to mounting height and angle; the 10cm and 20–30 degree parameters came from physical testing and are not arbitrary, so locking them in early matters for the mechanical design.
- I need to start drafting the final testing algorithm code this week. The CT demo code is a solid base but it has no ball pickup or sorting logic, no servo control, and no IR-guided fine positioning. There is significant work to do before the robot can handle the full R→G→B cycle with deposits at the correct bases.
- I communicated the LCD test results, the PixyCam positioning parameters, and the HD evaluation chart to the team during the meeting. Everyone left with a clear picture of what needed to happen before Thursday.

3.4.2 Entry 12 (09/04)

What am I doing well?

- Getting the final testing code written before assembly was important. Writing it now means we can upload and test the moment the robot is wired rather than scrambling to write code while everyone is waiting around. The `Config` namespace approach was deliberate; keeping all the tunable constants in one place means that when we need to adjust speeds or thresholds during testing, it is one section to change rather than hunting through the whole file. The 11-step `runSequence()` structure also makes the logic easy to follow and debug, which matters when something goes wrong during a test run and you need to isolate which step failed.

What is our team doing well?

- The Week 9 timeline we established is realistic and accounts for the dependencies between subgroups. The fact that Franklin cannot finalise the CAD until Yusuf completes the circuit diagram is a real dependency, and calling it out explicitly means neither subgroup gets blindsided. Setting hard deadlines rather than loose targets also helps; in the past we have set soft goals that slipped without consequence. Code being done by Tuesday-Wednesday is non-negotiable because the mechanical subgroup needs to know where components sit and we need time to test.

Why were the task(s) important?

- The final testing code is the foundation that everything else depends on. The mechanical subgroup needs to know where the servos, sensors, and Arduino will sit. The electrical subgroup needs to know which pins are in use before they can finalise the circuit diagram. Writing the code first and sharing the pin assignments and timing requirements gives both subgroups what they need to complete their work. Without it, they would be designing around unknowns. The Week 9 timeline also matters because we have very little time between assembly and the final demonstration; any delays in Week 9 directly compress our testing window.

What could I have done better?

- I wrote the entire `final_testing.ino` before sharing it with anyone, which meant the rest of the team had no visibility into the logic until the Thursday meeting. Jesse and Yusuf in particular have useful perspectives on what the robot will physically be able to do, and getting their input earlier might have caught assumptions I made about sensor timing or servo behaviour that do not hold in practice. Even a rough draft shared on Monday would have been better than a completed file presented on Thursday.

Is the progress in alignment with schedule?

- Code is done, the wheel mould is printing, and the LCD mount is printed. The Week 9 timeline is set. On paper we are on track, but the timeline is tight; if the 3D prints have issues or the circuit diagram takes longer than expected, we lose testing time that we cannot recover. I am more cautious than the schedule suggests because a lot of things need to go right in sequence next week.

What have I learned and what do I need to learn?

- I learned how to structure a complex multi-step robot algorithm in a way that is both readable and testable. Using a namespace for configuration constants and breaking the sequence into named step functions makes the code much easier to debug than a monolithic loop. I also learned that writing out the full 11-step sequence forced me to think carefully about edge cases I had not considered, like what happens if the ball is not in the clamp after a pickup attempt, or if the base scan times out.
- I need to be ready for significant code changes once we start testing on hardware. The timing values in `Config` are estimates; the actual turn durations, approach speeds, and servo angles will all need calibration runs before they are reliable. I also need to test the bus isolation functions for the ultrasonic sensor under real conditions, since I2C and SPI interference has not been verified on the new chassis yet.
- I communicated the full code structure and the Week 9 timeline to the team at the Thursday meeting so everyone understood the sequence and their dependencies going into the final week.

3.5 Week 9

3.5.1 Entry 13 (13/04)

What am I doing well?

- I presented the algorithm and code clearly enough that the team understood how the 11-step sequence works. Walking through the draft code line by line meant everyone knew what to expect when testing starts.

What is our team doing well?

- The subgroup leader brainstorming caught potential issues before they happened: pinout routing, servo bracket placement, sensor conflicts. This kind of pre-build coordination prevents rework later.

Why were the task(s) important?

- Mechanical assembly is the foundation; everything else depends on it being done correctly. Making sure the team understands the algorithm before testing saves time during debugging sessions.

What could I have done better?

- We were behind schedule; the Week 9 timeline had assembly starting earlier. Better time management would have given us more buffer for electrical work.

Is the progress in alignment with schedule?

- No, we are running late. Assembly should have started earlier in the week. We need to catch up on Wednesday with electrical connections.

What have I learned and what do I need to learn?

- I learned that presenting code to non-programmers requires slowing down and explaining the logic, not just the syntax. The algorithm walkthrough helped the mechanical team understand what the robot will do.
- I need to be ready for debugging once the robot is wired; electrical issues are harder to diagnose than code issues.
- I communicated the algorithm structure and soldering plan to the team so everyone knows the sequence.

3.5.2 Entry 14 (15/04)

What am I doing well?

- The common GND solution for PWM noise filtering was important; without it the servos would jitter. Testing each connection with the multimeter caught the LCD VDD issue before it caused problems.

What is our team doing well?

- Yusuf and I worked efficiently on soldering while Franklin handled mechanical. Parallel work kept things moving.

Why were the task(s) important?

- Proper soldering and wiring is critical; a bad connection would cause intermittent failures during testing that are hard to debug.

What could I have done better?

- We should have tested the LCD pins before soldering. The VDD fault wasted time desoldering.

Is the progress in alignment with schedule?

- Yes, we are catching up. Wiring is done and the robot is ready for testing.

What have I learned and what do I need to learn?

- I learned the importance of common GND for PWM noise filtering and using the multimeter beep test for continuity.
- I need to test the full system with code now that everything is wired.

3.5.3 Entry 15 (17/04)

What am I doing well?

- Testing each component individually before combining them was the right approach. The servo jitter issue is a good example of why; if we had gone straight to a full system test, that fault would have been much harder to isolate because it could have looked like a code issue, a power issue, or a mechanical problem. By testing the servo alone first and ruling out the other variables, I was able to narrow it down to the solder joint quickly. The same logic applied to the IR and ultrasonic; confirming each works in isolation before combining gives you a clean baseline and makes debugging the integrated system faster.

What is our team doing well?

- The role split today was efficient. Jesse, Yusuf, and I handled the technical build while Franklin, Bradman, and Yizhuo kept making progress on the report and risk assessment. Nobody was idle and nothing blocked anyone else. Jesse getting the Pixy2 signatures trained in the same session means we are not scrambling to do that after integration, which would have eaten into testing time.

Why were the task(s) important?

- Individual component testing before full integration is critical for a system this complex. The robot has five major subsystems communicating over shared buses, and any one of them misbehaving can mask or cause symptoms in another. The servo jitter was caused by a wiring fault, not code, but without the individual test it could have wasted hours of debugging at the algorithm level. Confirming the IR calibration inside the clamp enclosure also matters because the clamp geometry means the sensor operates at very short range; its detection threshold in open air is different to its detection threshold when a ball is sitting inside the housing.
- Completing the signature training is also essential because the Pixy2's detection reliability depends on how well the signatures were trained for the specific lighting and distance conditions in the arena. Doing this on the assembled chassis rather than on a test bench means the training reflects the actual operational conditions.

What could I have done better?

- The servo solder fault could have been caught at the end of the 15th session if I had done a quick test before leaving rather than assuming the connection was fine. The joint looked solid visually but was not making reliable contact under load. Testing it briefly before packing up would have saved time fixing it at the start of this session.

Is the progress in alignment with schedule?

- We are slightly behind the Week 9 timeline that was set on the 9th; the plan had assembly and integration finishing by Tuesday-Wednesday, and we are doing this on Thursday. That said, all individual components are now confirmed working and the system is integrated, so the remaining task is end-to-end algorithm testing and refinement. Whether that is enough time depends on how many issues come up during full test runs.

What have I learned and what do I need to learn?

- I learned that a visually clean solder joint is not the same as a reliable one, especially on joints that carry PWM switching current. The servo battery negative terminal needed the multimeter to catch it, not a visual inspection. Reflowing a joint that looks fine is a good habit when a component behaves erratically.

- I also reinforced that bus isolation matters in practice and not just in theory. The fact that the ultrasonic worked cleanly with the I2C and SPI buses shut down confirmed the technique is working as intended on the real hardware.

- I need to run the full 11-step algorithm end-to-end on the actual robot as soon as possible. Each step has been tested or reasoned through individually, but the real test is whether they chain together cleanly, the state variables transfer correctly between steps, and the robot recovers gracefully if a ball is lost or a scan times out.

- I communicated the component test results and the servo fault diagnosis to the team. Jesse shared the trained Pixy2 signatures so everyone knows the robot's vision system is ready for testing.

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